SLA SENTINEL – INTELLIGENT SERVICE LEVEL AGREEMENT MONITORING AND VENDOR PERFORMANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "SLA Sentinel – Intelligent Service Level Agreement Monitoring and Vendor Performance Management System" is the bonafide work of Mohamed Abshar (240701317), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year , Semester V, under my supervision.

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DECLARATION

I, Mohamed Abshar (240701317), hereby declare that the project report titled “SLA Sentinel – Intelligent Service Level Agreement Monitoring and Vendor Performance Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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Mohamed Abshar

ABSTRACT

Organisations use SLAs to define vendor uptime, response and resolution targets, escalation procedures and service credits. Tracking these commitments through spreadsheets, e-mail and PDF contracts leads to missed deadlines, inconsistent downtime records and unclaimed credits.

SLA Sentinel centralises vendor performance management in Zoho Creator. Ten related forms capture agreements, incidents, downtime, escalations, credits and scorecards. Deluge scripts snapshot SLA targets, calculate deadlines and evaluate response and resolution independently. A 15-minute monitor supports severity-based escalation through Zoho Mail and Cliq. Monthly processing calculates uptime excluding planned maintenance, routes credits for approval and produces weighted scorecards. A Blueprint controls incident transitions; role-based permissions restrict vendor access; a REST integration provides external uptime data.

The report includes 18 functional and access-control test cases, deployment procedures and governance controls. The intended outcome is timely breach detection and auditable vendor accountability; live test results require confirmation as noted in Chapter 6.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Service Level Agreement Monitoring*

PROJECT AT A GLANCE

Attribute	Details
Project Title	SLA Sentinel – Intelligent Service Level Agreement Monitoring and Vendor Performance Management System
Application Domain	IT Service Management / Vendor & Contract Management
Student Name & Register No.	Mohamed Abshar – 240701317
Zoho Creator Edition	Educational Edition
Application Name (Link Name)	SLA Sentinel (sla-sentinel)
Published / Live URL	https://creatorapp.zohopublic.in/mohamedabshar.2024.cse_ra591/sla-sentinel-vendor-management-system/page-perma/DashBoard/OYDyF0YGzmJnQ12fwYZC5Za3VWGzg5mU2wbCr3yZ89CChApmSRmPsU588bPAQKR7MFFGBpUSWakzX1PbTQDUh6JkFXPbb6Kr9Awp
Demo Credentials (per role)	Admin: admin@sla-sentinel.demo / Admin@2026; Incident Manager: incident@sla-sentinel.demo / Incident@2026; Vendor Portal: vendor@sla-sentinel.demo / Vendor@2026
Number of Forms / Subforms	10 / 2
Number of Reports (by type)	List: 7 Kanban: 1 Calendar: 1 Timeline: 1 Pivot/Chart: 3 (total 13)
Number of Pages / Dashboards	3
Number of Workflows	Form: 5 Schedule: 4 Approval: 1 Blueprint: 1 Batch: 1
Deluge Functions / Custom Actions	13 scripts (4 functions) / 2 custom actions
Integrations	Zoho Mail; Zoho Cliq (OAuth); UptimeRobot REST API (invokeurl); Creator REST API v2
Roles & Profiles Defined	Administrator / SLA Manager, Incident Manager, Vendor Representative (Portal)
Environments Used	Development → Stage → Production
Date of Production Deployment	

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
ISP	Internet Service Provider
ITIL	Information Technology Infrastructure Library
ITSM	IT Service Management
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SLA	Service Level Agreement
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

External providers deliver services such as cloud hosting, connectivity, cybersecurity and IT support. Service Level Agreements (SLAs) define their performance obligations and financial consequences for non-compliance.

An SLA specifies availability, severity-based response and resolution targets, excluded maintenance, escalation procedures and service credits. For example, a contract may guarantee 99.9% monthly uptime and refund part of the monthly fee after a breach.

The project models a mid-sized organisation using multiple vendors. IT operations records incidents, an SLA manager maintains contracts, and management reviews vendor performance. Spreadsheet-based tracking makes these activities dependent on manual follow-up.

SLA Sentinel supports proactive monitoring through Zoho Creator forms, reports, dashboards, Deluge scripts, schedules, a Blueprint, approvals and integrations.

1.2 Problem Statement

Manual tracking separates incidents from contractual targets, delays breach detection and weakens uptime and credit calculations. The main problems are:

- **Manual tracking: staff must locate and calculate each incident's deadlines.**
- **Late detection: breaches may remain unnoticed until month-end review.**
- **Incomplete outage timestamps prevent reliable uptime measurement.**
- **Service credits are missed when contractual penalties are not calculated.**
- **Fragmented records prevent historical vendor comparisons.**

1.3 Objectives

- Design related forms for vendors, SLA agreements, incidents, downtime, escalations, service credits and scorecards.
- Automate deadlines, independent response/resolution evaluation and severity-based escalation, targeting monitoring intervals of 15 minutes.
- Calculate monthly uptime excluding planned maintenance and generate credits for approval.

- Provide role-based dashboards and weighted vendor scorecards.
- Integrate Zoho Mail, Cliq and an external uptime REST API.
- Support secure deployment, record-level permissions, audit history and governance.

1.4 Scope of the Project

In scope: vendor and agreement management; severity-based targets and credit tiers; incident registration and deadlines; Blueprint-controlled incident transitions; monitoring and escalation logs; downtime, uptime and approved credits; scorecards and dashboards. Administrator/SLA Manager, Incident Manager and Vendor Representative roles access the web application, mobile app or restricted portal.

Out of scope: direct network probing, business-hour and holiday calendars, automatic invoice deductions and multi-organisation operation. SLA clocks run continuously; external monitoring supplies uptime data. Chapter 9 lists extensions.

1.5 Justification for a Low-Code Approach

Zoho Creator suits this data-driven application: forms, lookups and subforms model records; reports and pages present results; workflows, schedules, Blueprints and approvals coordinate work; Deluge handles business rules. Platform authentication, hosting and portal access reduce custom development [1], [2].

The approach supports faster development, configurable SLA terms, integrated access controls and mobile delivery. Managers can maintain targets, credit tiers and escalation recipients as data rather than changing code [3].

1.6 Organisation of the Report

Chapters 2–4 cover the system study, requirements and design. Chapters 5–7 describe implementation, testing and deployment. Chapters 8–9 discuss results, limitations and future work.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing process (Figure 2.1), staff report disruptions by phone or e-mail and record incidents in a spreadsheet. A coordinator retrieves SLA targets from PDF contracts and follows up until restoration. Month-end downtime, credits and vendor reviews are calculated manually from incomplete records.

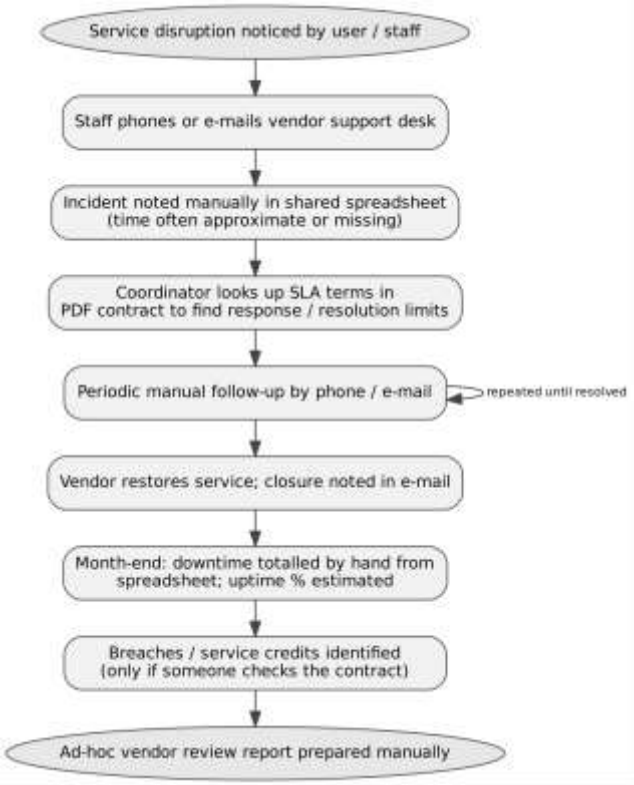


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- SLA targets live in PDF contracts, so the correct deadline for each incident is not visible where the incident is tracked.
- No reminders exist before a deadline; breaches are found after they happen, typically during month-end review.
- Timestamps for acknowledgement, response and restoration are incomplete, so response and resolution performance cannot be measured separately.
- Downtime is recorded inconsistently and planned maintenance is not separated, making uptime calculations unreliable.

- Service credits are computed manually, if at all, and there is no approval trail for claims.
- There is no access control — every spreadsheet user can see and change every row — and no audit trail of changes.
- There is no historical, comparable view of vendor performance for renewal or penalty decisions.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a conventional web application stack with Zoho Creator, using estimates for one student developer.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	About 10–12 weeks (schema, APIs, UI, auth, scheduler, deployment)	About 3–4 weeks (forms, reports, pages, Deluge, workflows)
Skills required	Front-end, back-end, SQL, DevOps, security	Data modelling, Deluge scripting, platform configuration
Hosting & infrastructure	Cloud VM/containers, database, backups and SSL managed by the team	Fully hosted by Zoho; backups and SSL included
Mobile application	Separate responsive UI or native app	Generated automatically (Zoho Creator mobile app)
Security & access control	Custom authentication, RBAC and audit logging to be coded	Built-in users, roles, permission profiles, record criteria, audit trail
Maintenance & change	Code change, test and redeploy for every rule change	SLA targets, credit tiers and recipients edited as data; scripts edited in browser
Customisation limits	Unlimited	UI layout and real-time (< 15 min) scheduling constrained by the platform
Cost	Hosting, database and developer time	Per-user subscription; education/free plan for development

2.4 Literature Survey

Table 2.2 reviews low-code research [4]–[6], SLA management sources [7], [8] and a commercial reference [9].

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Sanchis et al., [4]	Low-code as enabler of digital transformation in manufacturing industry	Low-code platforms let domain users build process applications quickly and integrate them with existing systems.	Supports building SLA tracking as a citizen-developer application owned by IT operations.
2	Bock & Frank, [5]	Low-code platform	Characterises low-code platforms by their data-model, process and UI generators, and notes limits in flexibility and vendor lock-in.	Guided which parts to model visually and informed the limitations discussed in Section 8.4.
3	Di Ruscio et al., [6]	Low-code development and model-driven engineering: two sides of the same coin?	Low-code is a practical form of model-driven engineering: models (forms, workflows) are executed directly by the platform.	Justifies treating forms, Blueprint and schedules as the executable design of the system.
4	Keller & Ludwig, [7]	The WSLA framework: specifying and monitoring SLAs for web services	SLAs need machine-readable parameters, measurement, evaluation against obligations and actions on violation.	Inspired the separation of targets (agreement), measurement (timestamps/downtime), evaluation (scripts) and action (escalation, credits).
5	AXELOS, [8]	ITIL 4 – Service Level Management practice	Defines SLA targets, monitoring, reporting, escalation and continual review of service levels.	Basis for the incident lifecycle, severity model and separate response/resolution measures.
6	ServiceNow [9]	Service Level Management (commercial ITSM product)	SLA definitions attach timers to tasks, with warning thresholds, breach flags and escalations.	Benchmark for features (SLA snapshot, at-risk warnings) at a fraction of the cost and complexity.

2.5 Emerging Trends in Low-Code Development

The design applies configurable business data, workflow automation, AI-assisted categorisation and API integration. Managers maintain contractual parameters; workflows connect monitoring, credits and approvals; Zia keywords support incident analysis; REST interfaces connect external monitoring systems [3].

2.6 Proposed System

SLA Sentinel addresses the existing system's limitations through the following capabilities:

- Snapshot agreement targets onto incidents so later contract edits do not change recorded deadlines.
- Monitor every 15 minutes; warn the vendor, notify the Incident Manager on breach, escalate continuing breaches to the Administrator and retain an audit record.
- A **Blueprint** stamps acknowledgement, response and resolution times automatically, and response and resolution SLAs are evaluated **independently**.
- Downtime is recorded per agreement with planned maintenance flagged, and a **monthly schedule** computes achieved uptime.
- **Service-credit records** are generated from tiered rules and routed through an **approval process**.
- **Role-based access**, record-level criteria for vendor users and the **audit trail** replace the shared spreadsheet.
- **Scorecards, charts and dashboards** provide a comparable history of every vendor's performance.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator / SLA Manager	Owns vendor contracts and the SLA monitoring process	Register vendors; create agreements, targets and credit rules; configure escalation policy; review breaches; approve service credits; analyse scorecards; manage users
IT Operations / Incident Manager	Handles operational incidents day to day	Register incidents; assign personnel; move incidents through the lifecycle; record downtime; monitor pending and overdue incidents; receive Stage 2 alerts
Vendor Representative (portal user)	Employee of a vendor, external to the organisation	View only own vendor's incidents; acknowledge; update progress; submit resolution details; view own SLA performance
Management (indirect)	Senior IT / finance leadership	Consume dashboards and scorecards shared by the Administrator; no data entry

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow the Administrator to register vendors with a unique auto-generated Vendor ID, category, contacts, contract dates and status.	Admin	High
FR-02	The system shall allow the Administrator to create SLA agreements per vendor with guaranteed uptime, maintenance window, monthly fee, per-severity response/resolution targets and tiered credit rules.	Admin	High
FR-03	The system shall allow the Incident Manager to register an incident and shall offer only the active agreements of the selected vendor.	Incident Mgr	High
FR-04	The system shall reject incidents whose agreement does not belong to the vendor, is not active on the reported date, or whose detected time is after the reported time.	Incident Mgr	High

FR ID	Requirement	Role(s)	Priority
FR-05	On saving an incident, the system shall copy the applicable targets onto the incident and compute the response and resolution deadlines.	System	High
FR-06	The system shall control the incident lifecycle (Reported, Acknowledged, In Progress, On Hold, Escalated, Resolved, Reopened, Closed) and stamp acknowledgement, response and resolution times automatically.	Incident Mgr, Vendor	High
FR-07	The system shall evaluate the response SLA and the resolution SLA independently as Pending, At Risk, Met or Breached.	System	High
FR-08	The system shall check open incidents at least every 15 minutes and send Stage 1 (warning), Stage 2 (breach) and Stage 3 (continued breach) escalations exactly once each.	System	High
FR-09	The system shall log every escalation with incident, stage, breach type, recipient, reason and timestamp.	System	High
FR-10	The system shall record downtime per agreement, compute its duration, reject overlapping records and exclude planned maintenance from uptime.	Incident Mgr	High
FR-11	The system shall compute each agreement's achieved uptime monthly and generate service-credit records from the credit rules.	System	High
FR-12	The system shall route service credits to the Administrator for approval and notify the vendor of the decision.	Admin	Medium
FR-13	The system shall generate a monthly scorecard per vendor with compliance percentages, uptime, breaches, credits, score and rating.	System	Medium
FR-14	The system shall display role-specific dashboards and reports (list, kanban, calendar, timeline, chart, pivot).	All	Medium
FR-15	The system shall restrict vendor representatives to records of their own vendor.	Vendor	High
FR-16	The system shall post Critical-severity breaches to a Zoho Cliq channel and pull external uptime ratios from an uptime-monitoring API.	System	Medium
FR-17	The system shall mark lapsed agreements as Expired and warn the Administrator 30 days before an agreement expires.	System	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	List reports with up to 5,000 incident records shall load within 3 seconds; one monitor run over 200 open incidents shall finish within the Deluge execution limits.
NFR-02	Security	Users shall access only the forms, reports and records permitted by their role; vendor users shall see only their own vendor's data.
NFR-03	Usability	The application shall be usable on desktop browsers and the Zoho Creator mobile app; any incident shall be loggable in under one minute.
NFR-04	Availability	The application shall rely on Zoho's hosted infrastructure; breach detection latency shall not exceed 15 minutes.
NFR-05	Maintainability	All contractual parameters (targets, credit tiers, thresholds, recipients) shall be editable as data, without changing scripts.
NFR-06	Auditability	All record changes shall be traceable through the audit trail; escalations and credit decisions shall be stored as records.
NFR-07	Data integrity	SLA terms applied to an incident shall not change when the agreement is later edited.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [Education / Professional] Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Mail (notifications), Zoho Cliq (critical alerts channel)
External service	UptimeRobot monitoring API (v2) – free plan
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 models three user roles and the scheduler. Tables 3.5–3.6 detail incident registration and deadline monitoring.

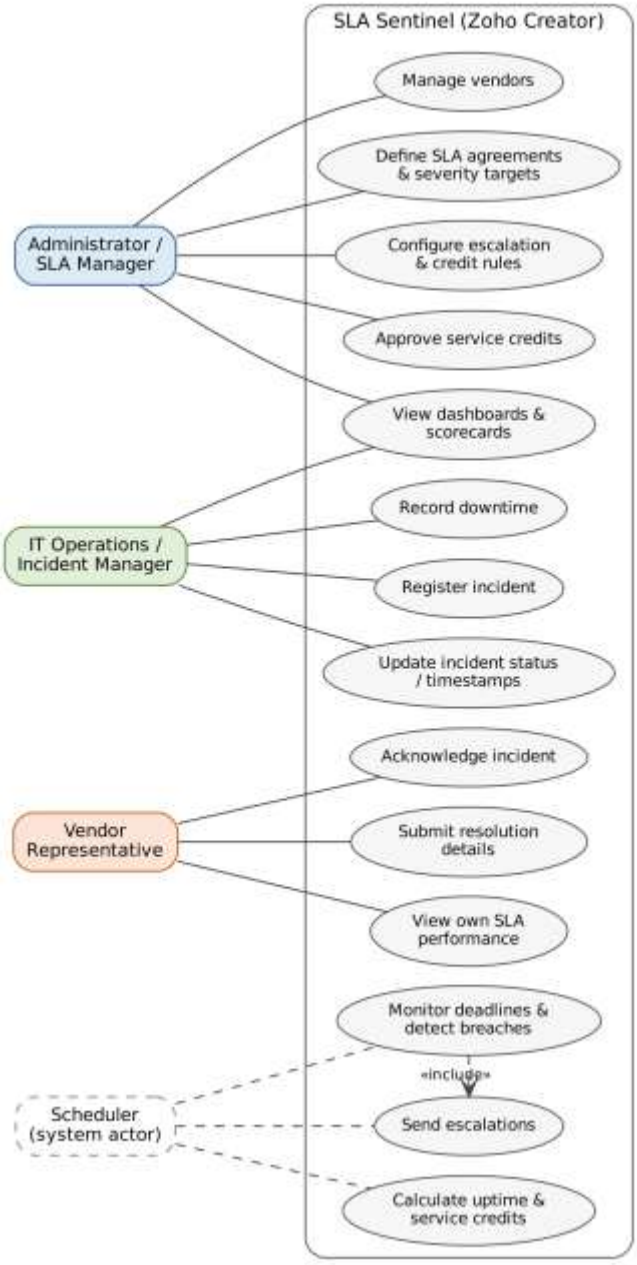


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Register Incident

Use Case ID	UC-01
Use Case Name	Register Incident
Actor(s)	Incident Manager (primary); Vendor Representative (notified)
Pre-condition	Vendor and an active SLA agreement with targets for the chosen severity exist; user is logged in as Incident Manager or Administrator.

Main Flow	1. User opens Report Incident. 2. Selects Vendor; the agreement list is filtered to that vendor's active agreements. 3. Enters title, description, category, severity, detected and reported times; ticks Service Outage if applicable. 4. Submits; validations run. 5. System copies targets, computes both deadlines, sets SLA statuses to Pending and e-mails the vendor.
Alternate Flow	4a. Agreement not of this vendor / not active / detected after reported → error shown, record not saved. 5a. No target for this severity → statuses set to Not Defined and Administrator alerted.
Post-condition	Incident stored in Reported stage with deadlines; it appears on Open Incidents, the Kanban board and the deadline calendar.
Related FR	FR-03, FR-04, FR-05

Table 3.6: Use Case Description – Monitor Deadlines and Escalate

Use Case ID	UC-02
Use Case Name	Monitor Deadlines and Escalate
Actor(s)	Scheduler (system); recipients: Vendor Representative, Incident Manager, SLA Administrator
Pre-condition	Open incidents with computed deadlines exist; Scheduler_Control record SLA_MONITOR holds the next run time.
Main Flow	1. Schedule fires. 2. System loads escalation policies and open incidents. 3. For each incident, compares elapsed time with warning threshold and deadlines. 4. Updates SLA status and escalation stage; calls logEscalation for each new stage. 5. Re-arms the schedule 15 minutes later.
Alternate Flow	3a. Status already At Risk/Breached for that stage → no duplicate notification. 4a. Critical severity at Stage ≥ 2 → message also posted to Zoho Cliq.
Post-condition	SLA statuses current; Escalation_Log contains one record per stage reached.
Related FR	FR-07, FR-08, FR-09, FR-16

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

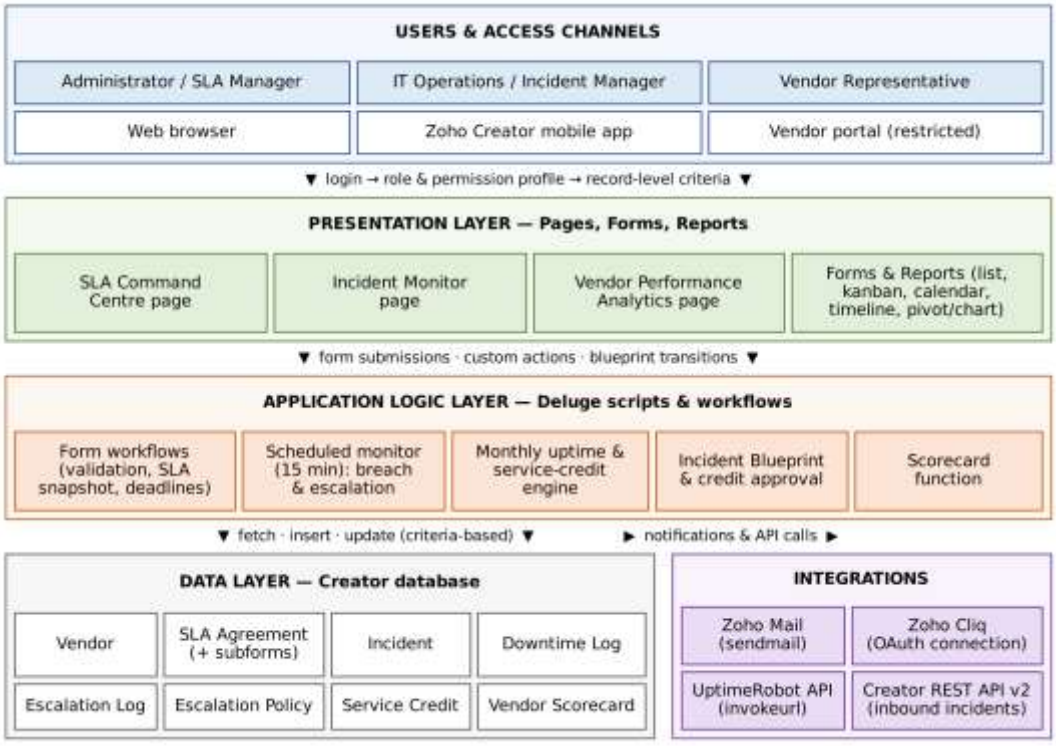


Figure 4.1: System Architecture of the Application

Figure 4.1 separates access, presentation, logic and data. Role permissions and vendor criteria control web, mobile and portal access. Forms, reports and UI Builder pages present records; Deluge, schedules, the incident Blueprint and credit approvals apply business rules. Ten related forms store data. Zoho Mail, Cliq, UptimeRobot and Creator REST interfaces support communication and integration.

4.2 Data Model

Figure 4.2 and Table 4.1 define the relationships. SLA_Targets and Credit_Rules are agreement subforms; other entities use lookups for independent reporting.

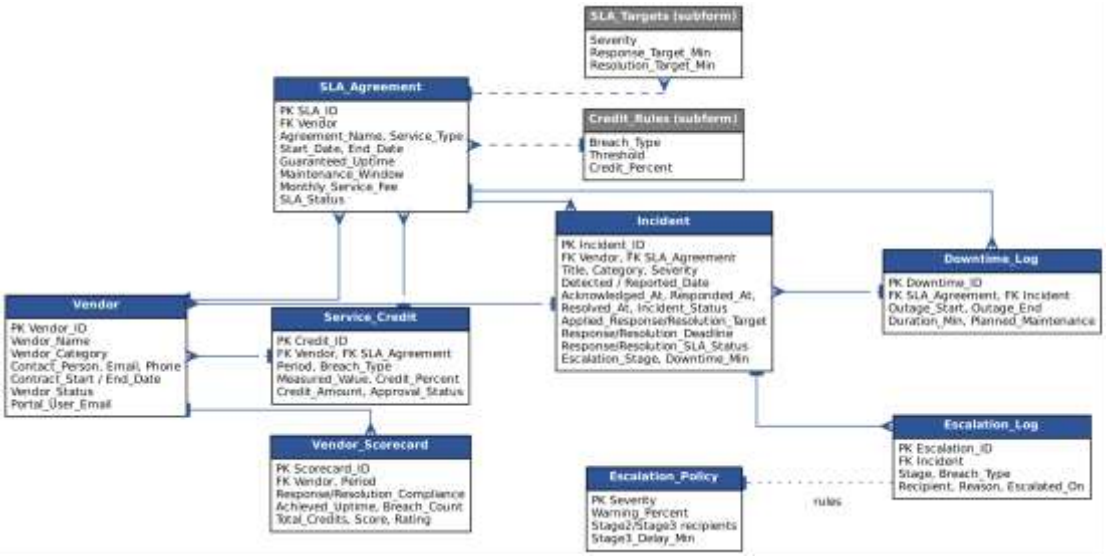


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Vendor	Vendor_ID, Vendor_Name, Email	SLA_Agreement	One-to-Many	Lookup in SLA_Agreement
SLA_Agreement	SLA_ID, Guaranteed_Uptime	SLA_Targets	Parent–Child	Subform
SLA_Agreement	SLA_ID	Credit_Rules	Parent–Child	Subform
Vendor / SLA_Agreement	Vendor_ID / SLA_ID	Incident	One-to-Many	Lookups in Incident
Incident	Incident_ID	Escalation_Log	One-to-Many	Lookup in Escalation_Log
Incident	Incident_ID	Incident_Update	One-to-Many	Lookup in Incident_Update
Incident / SLA_Agreement	Incident_ID / SLA_ID	Downtime_Log	One-to-Many	Lookups in Downtime_Log
Vendor / SLA_Agreement	Vendor_ID / SLA_ID	Service_Credit	One-to-Many	Lookups in Service_Credit
Vendor	Vendor_ID	Vendor_Scorecard	One-to-Many (one per period)	Lookup in Vendor_Scorecard
Escalation_Policy	Severity	Incident (by severity)	Reference (rule)	Matched by Severity in Deluge
Scheduler_Control	Job_Name	—	Configuration	Date-time field drives schedule

4.3 Form Design

Tables 4.2–4.6 specify the main forms using Zoho Creator field types.

Table 4.2: Field Specification – Vendor Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Vendor ID	Vendor_ID	Auto Number	—	Prefix VEN-	System generated
Vendor Name	Vendor_Name	Single Line	Yes	No duplicate values	Registered name
Vendor Category	Vendor_Category	Drop Down	Yes	—	Cloud, ISP, Cybersecurity, Software Maint., IT Support, Data Centre
Contact Person	Contact_Person	Name	Yes	—	
Email	Email	Email	Yes	Valid e-mail	Stage 1 recipient
Phone Number	Phone_Number	Phone	No	—	
Contract Start Date	Contract_Start_Date	Date	Yes	—	
Contract End Date	Contract_End_Date	Date	Yes	After start date	
Vendor Status	Vendor_Status	Radio	Yes	Default: Active	Active / Inactive / Suspended
Portal User Email	Portal_User_Email	Email	No	—	Used by portal record criteria
Description	Description	Multi Line	No	—	

Table 4.3: Field Specification – SLA Agreement Form (with subforms)

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
SLA ID	SLA_ID	Auto Number	—	Prefix SLA-	
Vendor	Vendor	Lookup (Vendor)	Yes	Active vendors only	
Agreement Name	Agreement_Name	Single Line	Yes	—	
Service Type	Service_Type	Drop Down	Yes	—	Hosting, Connectivity, Security, …
Start / End Date	Start_Date, End_Date	Date	Yes	End after start	
Guaranteed Uptime (%)	Guaranteed_Uptime	Decimal	Yes	90.000–100.000	e.g. 99.900

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Maintenance Window	Maintenance_Window	Single Line	No	—	e.g. Sun 01:00–03:00
Monthly Service Fee	Monthly_Service_Fee	Currency (INR)	Yes	> 0	Base for credits
SLA Status	SLA_Status	Drop Down	Yes	Default: Active	Active / Expired / Suspended
Contract Document	Contract_Document	File Upload	No	PDF	
Monitor ID	Monitor_ID	Single Line	No	—	UptimeRobot monitor
Last Period / Uptime	Last_Period, Last_Period_Uptime	Single Line / Decimal	—	Set by DS-08	Read-only
External Uptime 30d	External_Uptime_30d	Decimal	—	Set by DS-13	Read-only
SLA Targets (subform)	SLA_Targets	Subform	Yes	One row per severity	Severity (Drop Down), Response_Target_Min, Resolution_Target_Min (Number)
Credit Rules (subform)	Credit_Rules	Subform	No	—	Breach_Type (Uptime / Resolution), Threshold (Decimal), Credit_Percent (Decimal)

Table 4.4: Field Specification – Incident Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Incident ID	Incident_ID	Auto Number	—	Prefix INC-	
Incident Title	Incident_Title	Single Line	Yes	Max 120 chars	
Description	Description	Multi Line	Yes	—	Zia keyword source
Vendor	Vendor	Lookup (Vendor)	Yes	—	DS-01 filters agreement
SLA Agreement	SLA_Agreement	Lookup	Yes	Validated by DS-02	
Incident Category	Incident_Category	Drop Down	Yes	—	Outage, Degradation, Security, Performance, Other

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Severity	Severity	Radio	Yes	Default: Medium	Critical / High / Medium / Low
Service Outage	Service_Outage	Decision Box	No	Default: unticked	Creates downtime on resolve
Detected / Reported Date	Detected_Date, Reported_Date	Date-Time	Reported: Yes	Reported default: now	
Acknowledged / Responded / Resolved At	Acknowledged_At, Responded_At, Resolved_At	Date-Time	No	Set by Blueprint	
Incident Status	Incident_Status	Drop Down	—	Default: Reported	Mirrors Blueprint stage
Assigned Personnel	Assigned_Personnel	Users	No	—	
Applied Response / Resolution Target (min)	Applied_Response_Target_Min, Applied_Resolution_Target_Min	Number	—	Set by DS-03	SLA snapshot (read-only)
Response / Resolution Deadline	Response_Deadline, Resolution_Deadline	Date-Time	—	Set by DS-03	Read-only
Response / Resolution SLA Status	Response_SLA_Status, Resolution_SLA_Status	Drop Down	—	Pending	At Risk / Met / Breached / Not Defined
Escalation Stage	Escalation_Stage	Number	—	0	0–3
Downtime (min), Resolution Duration (min)	Downtime_Min, Resolution_Duration_Min	Number	—	Set by DS-04	
Resolution Notes	Resolution_Notes	Multi Line	On Resolve	Blueprint mandatory	
Keywords	Keywords	AI field (Keyword Extraction)	—	Source: Description	Zia
Reopen Count, Last Checked	Reopen_Count, Last_Checked	Number / Date-Time	—	System	Hidden on add

Table 4.5: Field Specification – Downtime Log, Escalation Log, Service Credit, Incident Update and Vendor Scorecard Forms

Form	Field (Link Name)	Field Type	Mandatory	Validation / Default	Remarks
Downtime_Log	Downtime_ID; Vendor; SLA_Agreement; Incident	Auto Number (DTL-); Lookups	Yes (except Incident)	—	

Form	Field (Link Name)	Field Type	Mandatory	Validation / Default	Remarks
Downtime_Log	Outage_Start; Outage_End; Duration_Min; Planned_Maintenance	Date-Time; Number; Decision Box	Yes	End > start, no overlap (DS-07)	Planned excluded from uptime
Escalation_Log	Escalation_ID; Incident; Stage; Breach_Type; Recipient; Reason; Escalated_On	Auto Number (ESC-); Lookup; Number; Drop Down; Email; Multi Line; Date-Time	—	Created only by DS-05	Read-only for all roles
Service_Credit	Credit_ID; Vendor; SLA_Agreement; Period; Breach_Type; Measured_Value; Credit_Percent; Credit_Amount	Auto Number (CR-); Lookups; Single Line; Drop Down; Decimal; Decimal; Currency	—	Created by DS-08	
Service_Credit	Approval_Status; Approved_By; Decision_On; Remarks	Drop Down; Users; Date-Time; Multi Line	—	Pending Approval	Approval process
Incident_Update	Incident; Update_Note; Posted_By; Posted_On	Lookup; Multi Line; Users; Date-Time	Incident, Note	Posted_By = login user; Posted_On = now	Vendor progress notes
Vendor_Scorecard	Vendor; Period; Total_Incidents; Response_Compliance; Resolution_Compliance; Achieved_Uptime; Breach_Count; Total_Credits; Performance_Score; Rating	Lookup; Single Line; Number; Percent; Percent; Decimal; Number; Currency; Decimal; Drop Down	—	Created by DS-09	One per vendor per month

Table 4.6: Field Specification – Configuration Forms (Escalation Policy, Scheduler Control)

Form	Field (Link Name)	Field Type	Mandatory	Validation / Default	Remarks
Escalation_Policy	Severity	Drop Down	Yes	No duplicates	One row per severity
Escalation_Policy	Warning_Percent; Stage3_Delay_Min	Number	Yes	Default 75; 60	Critical: 50 %, 15 min
Escalation_Policy	Stage2_Recipient_Email; Stage3_Recipient_Email	Email	Yes	—	Incident Mgr; SLA Admin

Form	Field (Link Name)	Field Type	Mandatory	Validation / Default	Remarks
Scheduler_Control	Job_Name; Next_Run; Last_Run	Single Line; Date-Time; Date-Time	Yes	SLA_MONITOR	Next_Run drives DS-06

4.4 Process and Workflow Design

DS-06 monitors Pending, At Risk and Breached states and tracks escalation progress (Figure 4.3). Blueprint transitions stamp timestamps through DS-10; DS-04 uses them to evaluate response and resolution separately (Figure 4.4).

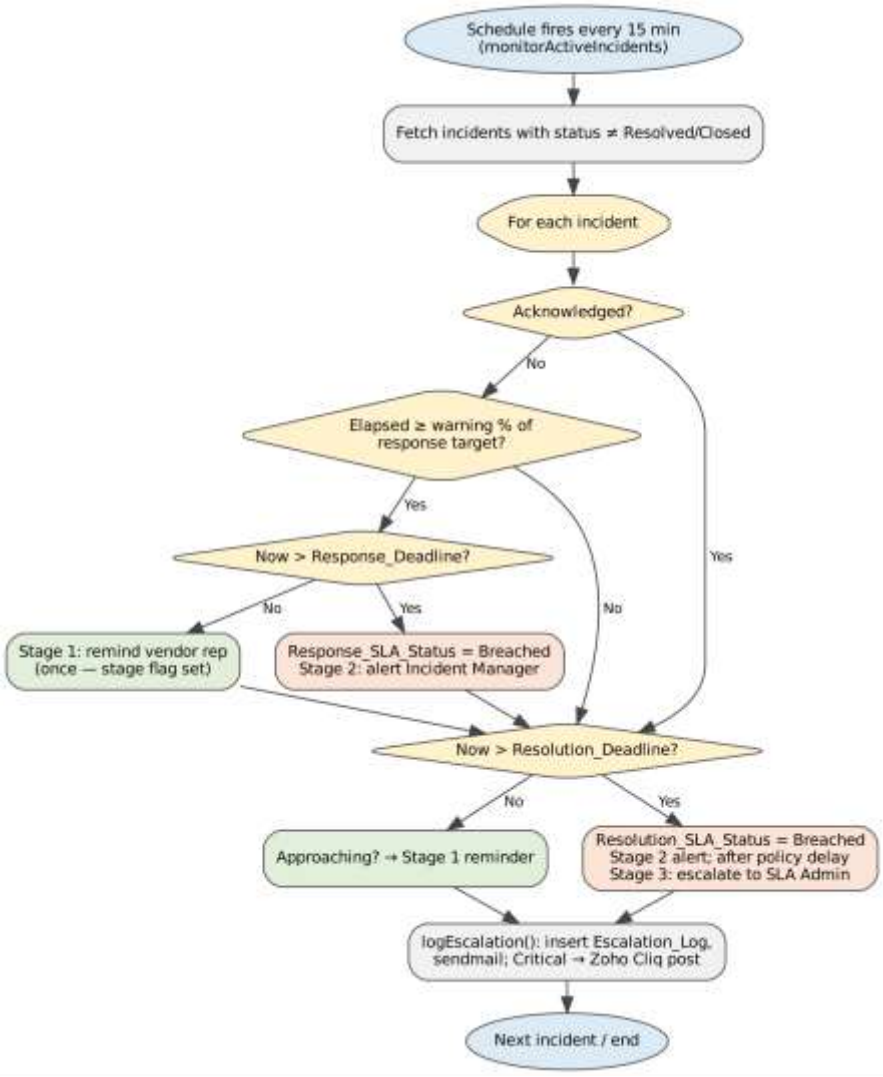


Figure 4.3: Workflow / Process Flow for SLA Deadline Monitoring and Escalation

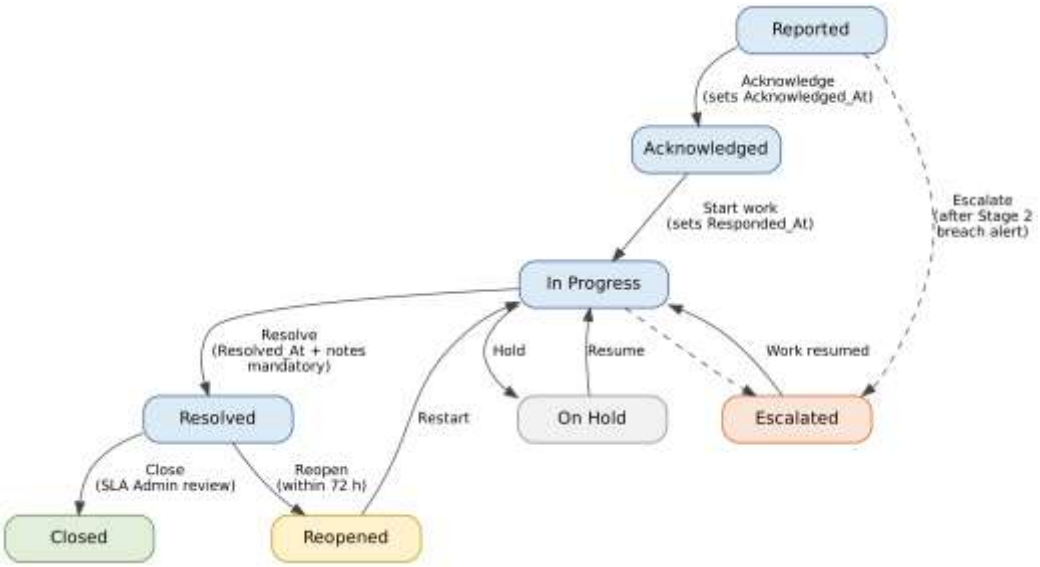


Figure 4.4: Incident Lifecycle (Blueprint) State Diagram

4.5 User Interface and Navigation Design

Figure 4.5 shows the navigation design. The dashboard wireframe (Figure 4.6) combines KPI tiles, charts, deadline-sorted incidents and quick actions.

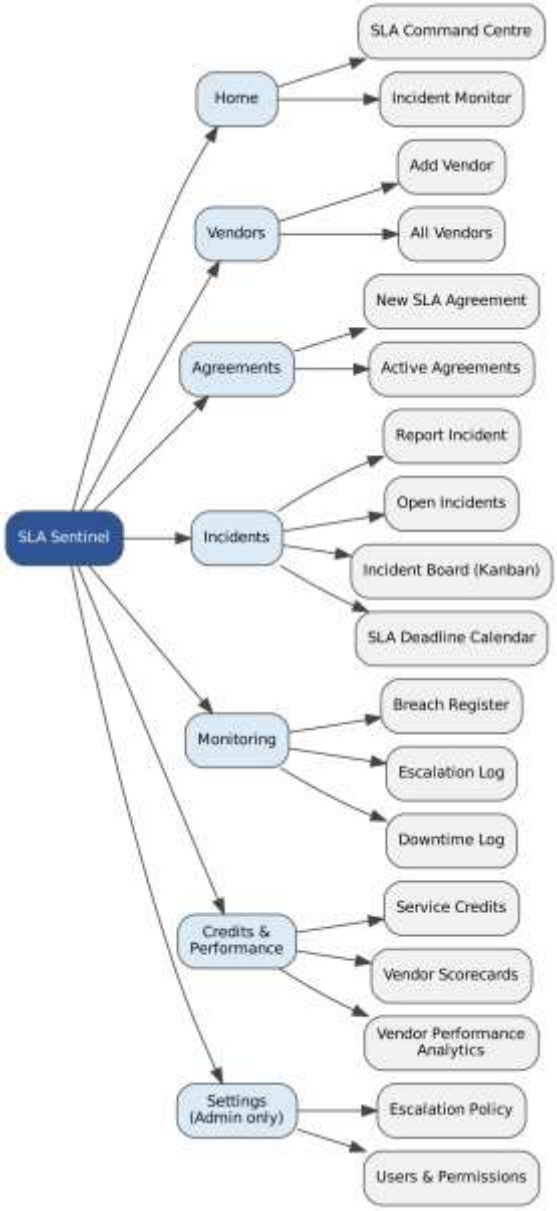


Figure 4.5: Navigation Structure of SLA Sentinel

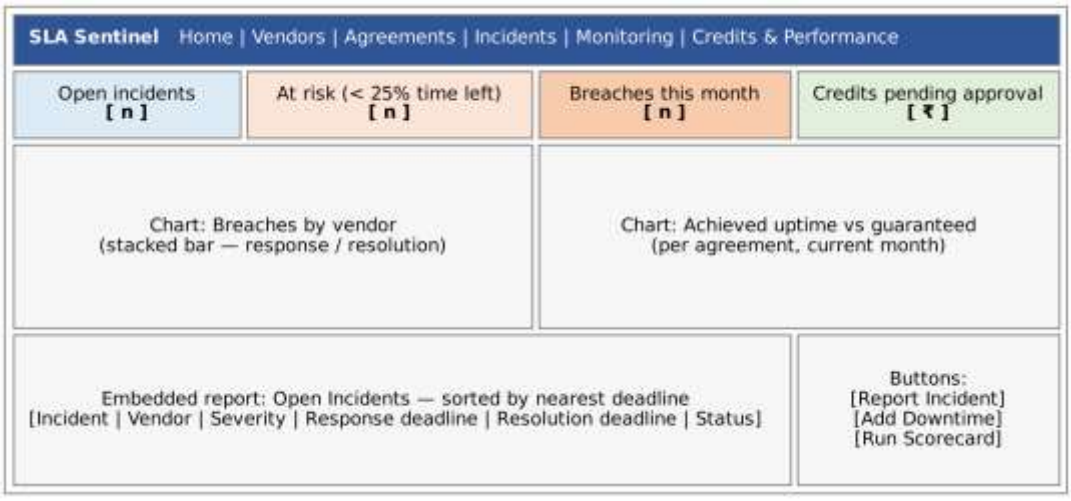


Figure 4.6: Wireframe of the SLA Command Centre Page

4.6 Access Control Design

Table 4.7: Role–Permission Matrix

Component	Administrator	Incident Manager	Vendor Rep. (Portal)
Form: Vendor	A E V D	V	V (own)
Form: SLA Agreement	A E V D	V	V (own)
Form: Incident	A E V D	A E V	E V (own; progress fields only)
Form: Downtime Log	A E V D	A E V	V (own)
Form: Escalation Log	V D	V	—
Form: Service Credit	E V (approve)	V	V (own, approved only)
Form: Vendor Scorecard	V D	V	V (own)
Forms: Escalation Policy, Scheduler Control	A E V D	—	—
Page: SLA Command Centre	V	V	—
Page: Incident Monitor	V	V	—
Page: Vendor Performance Analytics	V	V	—
Report: My Vendor Incidents (portal)	—	—	V

CHAPTER 5
IMPLEMENTATION

This chapter illustrates the dashboard, incidents, downtime, credits, rules and vendor performance. Chapter 7 shows activity auditing. Text, tables and code describe the remaining features.

5.1 Application Setup

The application was manually configured from the data model in Section 4.2, with role-based navigation, Administrator-only settings and a dashboard landing page. Figure 5.1 shows the application interface.

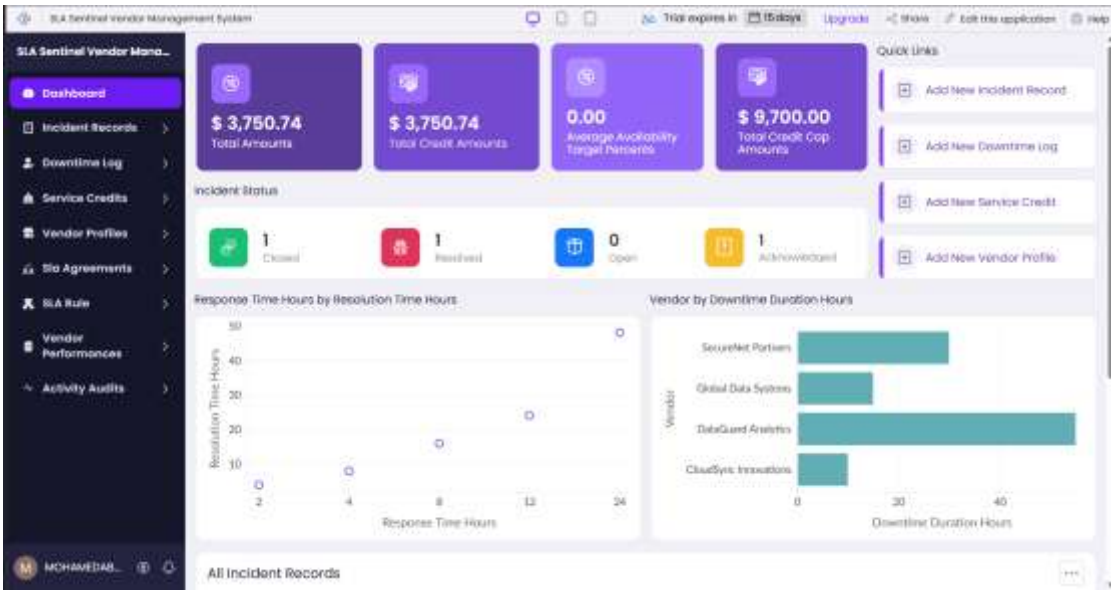


Figure 5.1: Application Dashboard and Navigation

5.2 Forms and Data Capture

The ten forms are Vendor, SLA_Agreement, Incident, Downtime_Log, Escalation_Log, Service_Credit, Vendor_Scorecard, Escalation_Policy, Scheduler_Control and Incident_Update. SLA_Agreement includes SLA_Targets and Credit_Rules subforms. Field specifications in Chapter 4 cover identifiers, text, dates, contacts, numbers, currency, choices, files, users, lookups and AI output.

Validation enforces unique vendors and severity policies, vendor-filtered active agreements (DS-01), valid ownership and timestamps (DS-02), and non-overlapping downtime (DS-07). Computed SLA fields are hidden on entry and read-only on edit. The success message displays the resolution deadline; SLA_Targets allows one row for each of four severities.

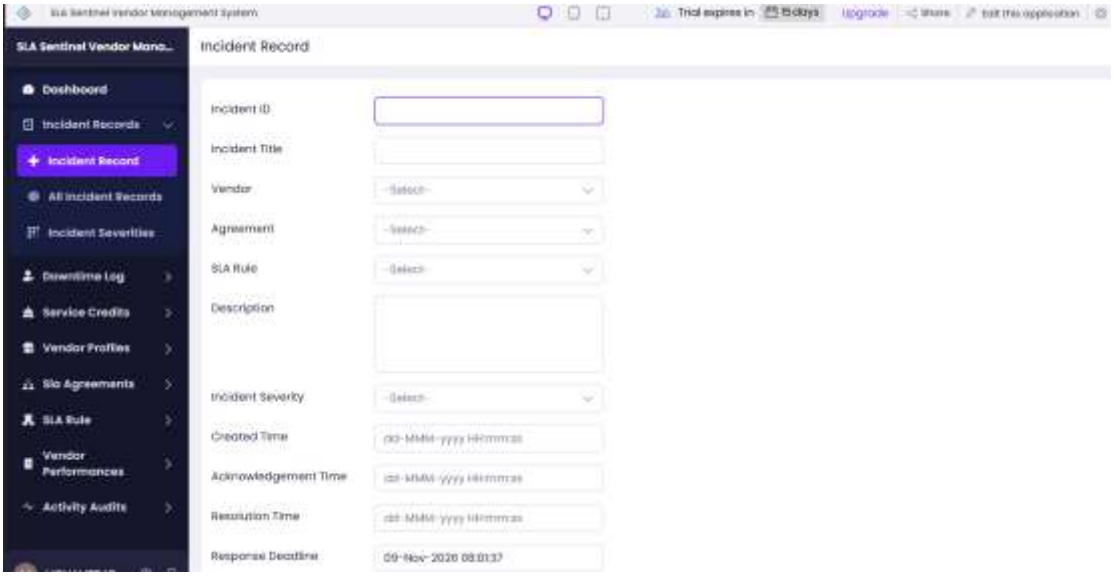


Figure 5.2: Incident Record Entry Form

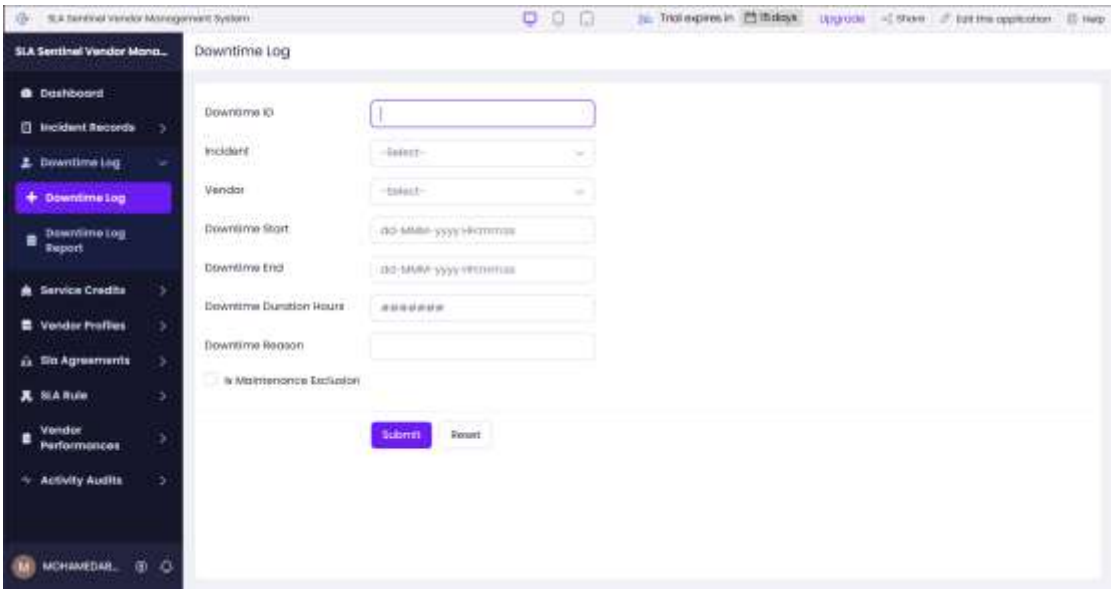


Figure 5.3: Downtime Log Entry Form

5.3 Reports

Table 5.1 lists reports and their purposes. Status colours distinguish Breached, At Risk and Met. Quick views show deadlines; detail views include related escalation and downtime records.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Open Incidents	List	Incident	Status not Resolved/Closed; group by Severity; sort by Resolution_Deadline asc	Recalculate SLA (DS-04); Add Downtime	Admin, Incident Mgr
Incident Board	Kanban	Incident	Columns by Incident_Status	Blueprint transitions	Admin, Incident Mgr
SLA Deadline Calendar	Calendar	Incident	Resolution_Deadline; colour by Severity	—	Admin, Incident Mgr
Incident Timeline	Timeline	Incident	Reported_Date → Resolved_At	—	Admin
Breach Register	List	Incident	Response or Resolution status = Breached; group by Vendor	Export PDF	Admin
Escalation Log	List	Escalation_Log	Sort by Escalated_On desc; group by Stage	—	Admin, Incident Mgr
Downtime Log	List	Downtime_Log	Group by SLA_Agreement; sum Duration_Min	—	Admin, Incident Mgr
Service Credits	List	Service_Credit	Group by Approval_Status; sum Credit_Amount	Approve / Reject	Admin
Vendor Scorecards	List	Vendor_Scorecard	Group by Period; sort by Performance_Score desc	—	Admin, Incident Mgr
Breaches by Vendor	Chart (stacked bar)	Incident	Count by Vendor × breach type	—	Admin
Uptime vs Target	Chart (bar + line)	SLA_Agreement	Last_Period_Uptime vs Guaranteed_Uptime	—	Admin
Incident Pivot	Pivot	Incident	Rows Vendor; columns Severity; values count, avg Resolution_Duration_Min	—	Admin
My Vendor Incidents	List	Incident	Portal record criteria (own vendor)	Blueprint transitions	Vendor Rep.

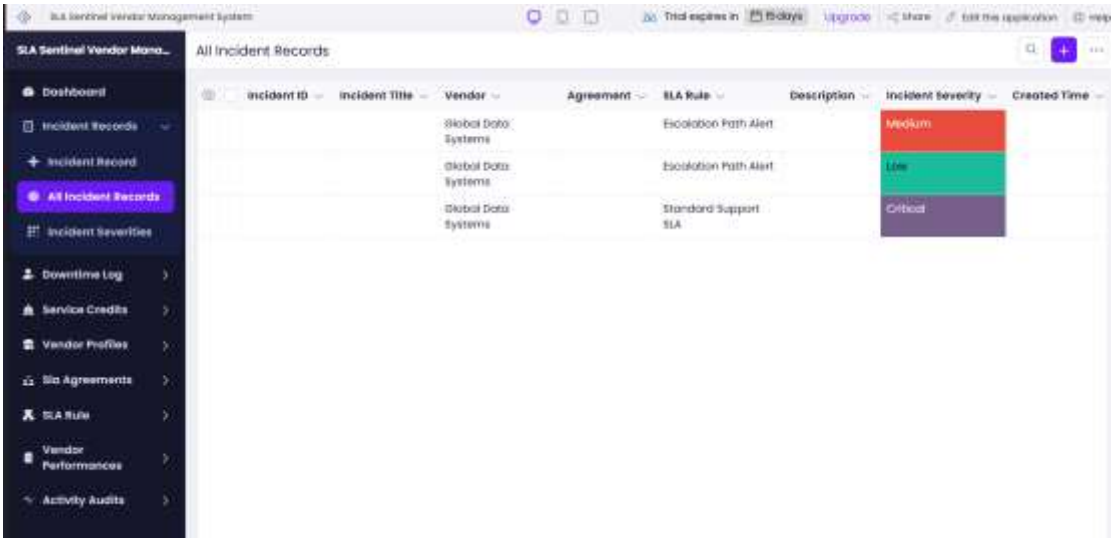


Figure 5.4: All Incident Records Report

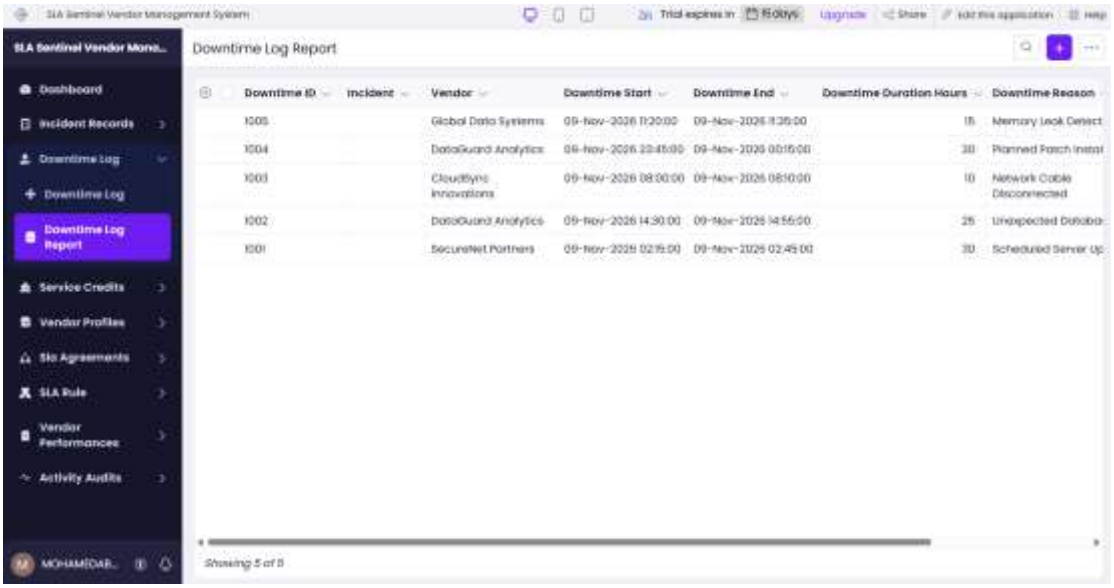


Figure 5.5: Downtime Log Report

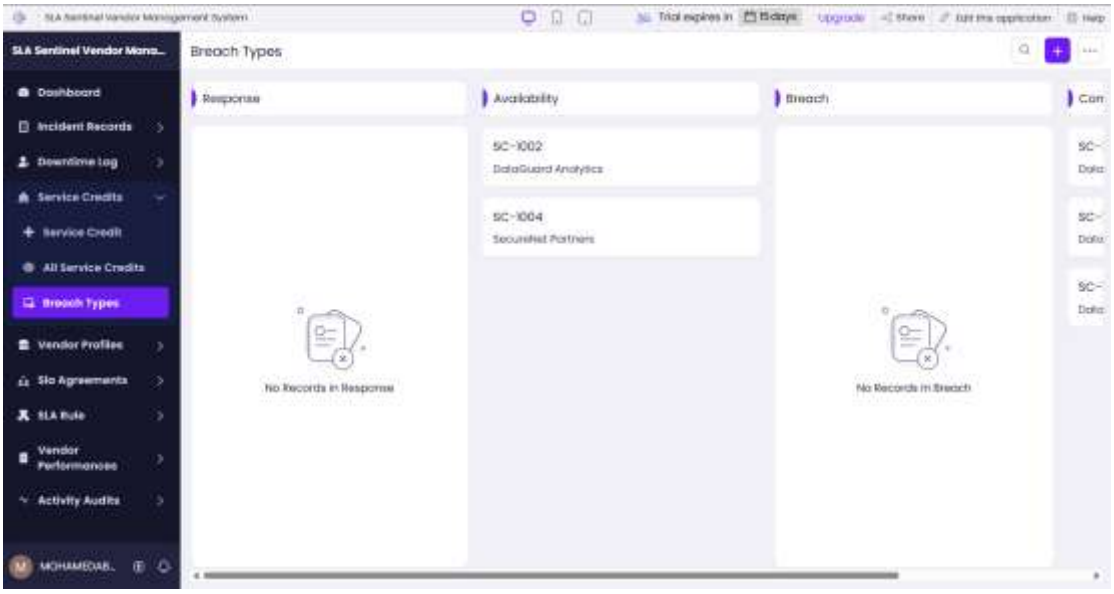


Figure 5.6: Service Credits Grouped by Breach Type

5.4 Pages and Dashboards

The three-page design comprises SLA Command Centre for KPIs, charts, open incidents and quick actions; Incident Monitor for a Kanban board, deadline calendar and severity filter; and Vendor Performance Analytics for scorecards, pivots and vendor-specific filtering.

5.5 Deluge Scripting

Table 5.2 summarises thirteen scripts. Appendix C retains the complete source, using Deluge record operations, collections and date functions [10].

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Incident → Vendor → On User Input	Filter agreements to the selected vendor	criteria fetch, getAll, ui.add, clear
DS-02	Incident → On Add → On Validate	Validate agreement, period and timestamps	fetch, if-else, alert, cancel submit
DS-03	Incident → On Add → On Success	SLA snapshot, deadlines, vendor e-mail	subform loop, addMinutes, sendmail
DS-04	Function evaluateIncidentSLA	Independent response/resolution evaluation; downtime record	function, list contains, ifnull, insert
DS-05	Function logEscalation	Log, e-mail and Cliq-post an escalation	insert, sendmail, Map, zoho.cliq
DS-06	Schedule: SLA Monitor (15 min)	Staged warnings and breaches	Map cache, for each, toLong, thisapp call

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-07	Downtime_Log → On Validate	Duration and overlap check	arithmetic, count with criteria
DS-08	Schedule: Monthly SLA Close	Uptime, credits, scorecards	date functions, nested loops, Map
DS-09	Function generateScorecard	Weighted score and rating	aggregation in loops, upsert
DS-10	Blueprint transitions	Stamp timestamps, re-evaluate SLA	zoho.currenttime, function call
DS-11	Approval: Service Credit	Record decision, notify vendor	zoho.loginuser, sendmail
DS-12	Schedule: Daily housekeeping	Expire agreements; expiry warning	date criteria, string building
DS-13	Function syncUptimeRobot	Pull external uptime ratios	invokeurl, connection, JSON parsing

Code 5.1: Computing deadlines from the SLA snapshot (extract of DS-03)

```
// DS-03 | Form: Incident | Event: On Add -> On Success (extract)
agr = SLA_Agreement[ID == input.SLA_Agreement];
for each t in agr.SLA_Targets
{
  if(t.Severity == input.Severity)
  {
    respTarget = t.Response_Target_Min;
    resTarget = t.Resolution_Target_Min;
  }
}
// targets are frozen on the incident
input.Applied_Response_Target_Min = respTarget;
input.Applied_Resolution_Target_Min = resTarget;
input.Response_Deadline = input.Reported_Date.addMinutes(respTarget);
input.Resolution_Deadline = input.Reported_Date.addMinutes(resTarget);
input.Response_SLA_Status = "Pending";
input.Resolution_SLA_Status = "Pending";
```

Code 5.2: Independent response and resolution evaluation (extract of DS-04)

```
// DS-04 | Function sla.evaluateIncidentSLA(int incId) (extract)
firstResponse = ifnull(inc.Acknowledged_At,inc.Responded_At);
if(firstResponse != null && openStates.contains(inc.Response_SLA_Status))
{
  if(firstResponse <= inc.Response_Deadline) { inc.Response_SLA_Status = "Met"; }
  else { inc.Response_SLA_Status = "Breached"; }
}
if(inc.Resolved_At != null && openStates.contains(inc.Resolution_SLA_Status))
{
  if(inc.Resolved_At <= inc.Resolution_Deadline) { inc.Resolution_SLA_Status =
"Met"; }
  else { inc.Resolution_SLA_Status = "Breached"; }
}
```

DS-03 snapshots severity targets and calculates deadlines from the reported time, preserving history after agreement edits (NFR-07). A 10:00 incident with 15-

minute response and four-hour resolution targets receives deadlines of 10:15 and 14:00.

DS-04 evaluates both obligations independently: acknowledgement at 10:10 and restoration at 15:00 meet the response target but breach resolution. Service outages generate a downtime record. DS-06 monitors escalation; DS-08 and DS-09 calculate monthly credits and scorecards.

5.6 Workflows and Automation

Table 5.3 lists the workflows. DS-06 uses a schedule attached to Scheduler_Control.Next_Run and advances that field by 15 minutes after each run [11].

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Agreement Filter	Form Workflow	Incident: Vendor on user input	—	Filter lookup	DS-01
Incident Validation	Form Workflow	Incident: On Add, On Validate	Rule violated	Block submission	DS-02
SLA Snapshot & Deadlines	Form Workflow	Incident: On Add, On Success	—	Compute deadlines, e-mail vendor	DS-03
SLA Re-evaluation	Form Workflow	Incident: On Edit, On Success	Timestamps changed	Call evaluateIncidentSLA	DS-04
Downtime Validation	Form Workflow	Downtime_Log: On Validate	—	Duration; overlap check	DS-07
Incident Lifecycle	Blueprint	Stage transitions	Role-restricted transitions	Stamp times; mandatory notes on Resolve	DS-10
SLA Monitor	Schedule (date-field, self-rescheduling)	Scheduler_Control.Next_Run	Open incidents	Warn / breach / escalate; re-arm	DS-06
Monthly SLA Close	Schedule	1st of month, 02:00	Active agreements	Uptime, credits, scorecards	DS-08
Agreement Housekeeping	Schedule	Daily, 08:00	End date	Expire; 30-day warning	DS-12
External Uptime Sync	Schedule	Daily, 06:00	—	Call syncUptimeRobot	DS-13
Service Credit Approval	Approval	Service_Credit created	Status = Pending Approval	Admin approves/rejects; vendor notified	DS-11

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Recalculate SLA	Batch / Report Workflow	Custom action on Open Incidents (multi-select)	Selected records	Re-run evaluateIncidentSLA	DS-04

Code 5.3: Staged escalation without duplicate notifications (extract of DS-06)

```
// DS-06 | Schedule: SLA Monitor (extract – resolution SLA)
if(now > inc.Resolution_Deadline)
{
  // Stage 2 - sent once
  if(inc.Resolution_SLA_Status != "Breached")
  {
    inc.Resolution_SLA_Status = "Breached";
    thisapp.sla.logEscalation(inc.ID,2,"Resolution","Resolution deadline
exceeded.");
  }
  else if(inc.Escalation_Stage < 3 && now.toLong() -
inc.Resolution_Deadline.toLong() >= stage3Delay * 60000)
  {
    thisapp.sla.logEscalation(inc.ID,3,"Resolution","Breach continuing -
escalated to SLA Administrator.");
    inc.Escalation_Stage = 3;       // Stage 3 - sent once
  }
}
else if(inc.Resolution_SLA_Status == "Pending" && elapsedMin >=
inc.Applied_Resolution_Target_Min * warnPct / 100)
{
  inc.Resolution_SLA_Status = "At Risk";       // Stage 1 - sent once
  thisapp.sla.logEscalation(inc.ID,1,"Resolution",warnPct + "% of resolution time
used.");
}
```

Escalation_Policy sets warning percentages, delays and recipients by severity. The example Critical policy warns at 50% and escalates after fifteen minutes of breach; Low warns at 90% and escalates after four hours. Escalation logs support monthly contractual review.

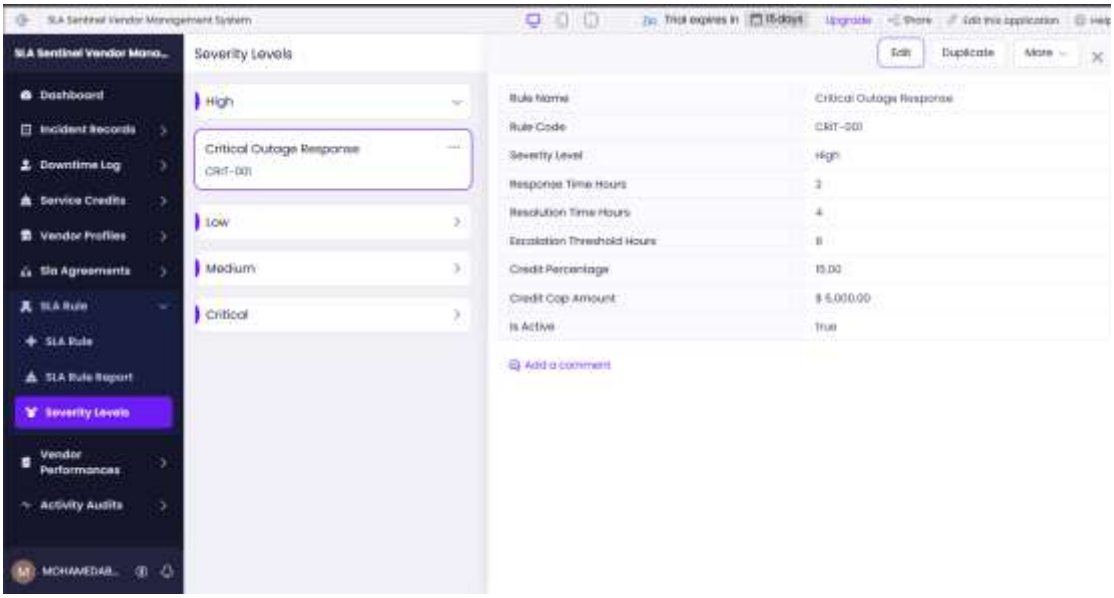


Figure 5.7: SLA Rule Details in the Severity Levels View

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
E-mail notifications	Zoho Mail	`sendmail` task	Platform (zoho.adminuserid)	New-incident, escalation, approval and expiry e-mails
Critical alerts	Zoho Cliq – channel slaalerts	Connection + `zoho.cliq.postToChannel`	OAuth 2.0 (connection cliq_sla_conn)	Stage ≥ 2 Critical breaches
External uptime	UptimeRobot API v2 – getMonitors [12]	`invokeurl` POST	API key held in connection uptimerobot_conn	30-day uptime ratio per monitor → External_Uptime_30d
Inbound incidents	Creator REST API v2 – Add Records (Incident form) [13]	REST POST from monitoring tools; tested in Postman	OAuth 2.0 (scope ZohoCreator.form.CREATE)	Monitoring tools can log incidents automatically; DS-02/DS-03 still run

Code 5.4: REST call through a secured connection (extract of DS-13)

```
// DS-13 | Function sla.syncUptimeRobot() (extract)
params = Map();
params.put("format","json");
params.put("custom_uptime_ratios","30");
response = invokeurl
[
  url :"https://api.uptimerobot.com/v2/getMonitors"
  type :POST
  parameters:params
  connection:"uptimerobot_conn"          // no API key in the script
];
for each m in response.get("monitors")
```

```
{
  for each a in SLA_Agreement[Monitor_ID == m.get("id").toString()]
  {
    a.External_Uptime_30d = m.get("custom_uptime_ratio").toDecimal();
  }
}
```

5.8 User Management and Role-Based Access

Table 4.7 defines three permission profiles. Administrators manage all components; Incident Managers edit operational records but only view agreements and credits. Vendor portal criteria restrict records to the signed-in vendor. Permitted Blueprint transitions and progress notes control incident updates; only Approved credits are visible to vendors.

5.9 AI and Analytics Features

Zia extracts keywords from incident descriptions for recurring-problem analysis. Keywords are compared with sample descriptions and do not determine SLA outcomes. Forms and scripts were prepared manually rather than through the AI application builder.

Charts and pivots compare breaches, uptime and resolution by vendor. Scorecards use 0.3 × response compliance + 0.4 × resolution compliance + 0.3 × uptime score. Ratings are Excellent (≥90), Good (≥75), Average (≥60) or Poor.

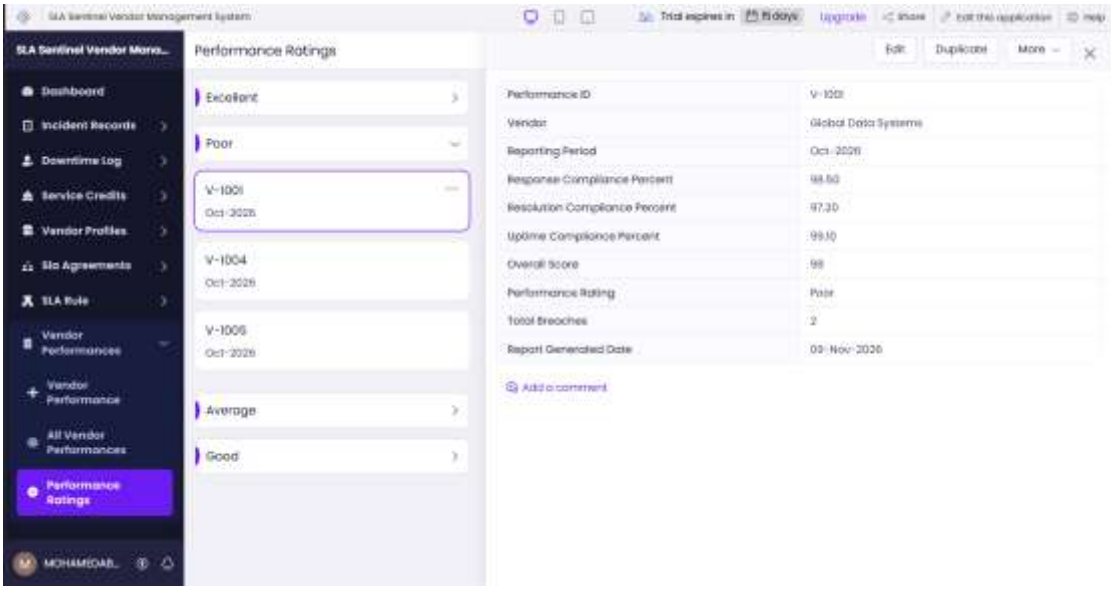


Figure 5.8: Vendor Performance Ratings and Compliance Details

5.10 Pro-Code Extension

No separate Catalyst function or JavaScript widget is included. REST APIs provide integration with external systems. Potential pro-code additions are a one-minute monitoring function and a live deadline-countdown widget.

5.11 Mobile and Portal Access

Mobile access uses the Zoho Creator app and Blueprint controls. The vendor portal limits navigation to the vendor's incidents, performance and approved credits.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The test strategy covers valid, invalid and boundary inputs; end-to-end incident, notification and credit workflows; role and vendor isolation; and user acceptance. Back-dated incident times exercise deadline behaviour. Table 6.1 records cases; Section 6.5 provides the acceptance-feedback template.

6.2 Test Cases

Table 6.1 lists the test cases. The *Actual Result* and *Status* columns are to be confirmed against the live application.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add vendor with all mandatory fields	CloudHost Solutions, Cloud, valid e-mail	Record saved, Vendor ID VEN-001	As expected	Pass
TC-02	FR-01	Add vendor with duplicate name	Vendor_Name = CloudHost Solutions again	Duplicate-value error shown	As expected	Pass
TC-03	FR-02	Create agreement with four severity targets and three uptime credit tiers	Critical 15/240 … Low 480/4320; tiers <99.9→5 %, <99→10 %, <95→25 %	Agreement saved with subform rows	As expected	Pass
TC-04	FR-03	Agreement lookup filtered by vendor	Select NetConnect in Incident form	Only NetConnect's active agreements listed	As expected	Pass
TC-05	FR-04	Agreement belonging to another vendor	Vendor NetConnect, agreement of CloudHost (via API)	Alert; record not saved	As expected	Pass
TC-06	FR-04	Detected time after reported time	Detected 10:30, Reported 10:00	Alert; record not saved	As expected	Pass
TC-07	FR-05	Deadline computation (Critical)	Reported 10:00, targets 15 / 240 min	Response 10:15, Resolution 14:00; statuses Pending	As expected	Pass
TC-08	FR-05, NFR-07	SLA snapshot preserved	Edit agreement Critical target to 30 min after TC-07	TC-07 incident deadlines unchanged	As expected	Pass
TC-09	FR-06, FR-07	Response met, resolution breached	Acknowledge 10:10, Resolve 15:00	Response = Met; Resolution = Breached	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-08	Stage 1 warning sent once	High incident, 80 % of response time elapsed; run monitor twice	Status At Risk; one Stage 1 log and e-mail only	As expected	Pass
TC-11	FR-08, FR-09	Stage 2 and Stage 3 escalation	Critical incident 20 min past resolution deadline, policy delay 15 min; run monitor twice	Stage 2 then Stage 3 logged; Cliq message posted	As expected	Pass
TC-12	FR-10	Overlapping downtime rejected	Second record 10:30–11:00 inside 10:00–12:00 for same agreement	Alert; record not saved	As expected	Pass
TC-13	FR-10, FR-11	Uptime excludes planned maintenance	60 min unplanned + 120 min planned in a 30-day month	Uptime = 99.861 %; credit tier 5 % created	As expected	Pass
TC-14	FR-12	Service credit approval	Administrator approves CR-001	Status Approved; Approved_By set; vendor e-mailed	As expected	Pass
TC-15	FR-13	Scorecard calculation	Resp. 90 %, Res. 80 %, uptime score 100	Score 89.0, Rating Good	As expected	Pass
TC-16	FR-15, NFR-02	Vendor isolation in portal	Login as SecureOps portal user	Only SecureOps incidents visible; NetConnect records not visible	As expected	Pass
TC-17	NFR-02	Incident Manager opens Escalation Policy	Direct URL to form	Access denied	As expected	Pass
TC-18	FR-17	Agreement expiry	Active agreement with End_Date yesterday; run housekeeping	Status Expired; new incidents against it rejected	As expected	Pass

6.3 Debugging Deluge Scripts

Debugging uses Deluge info statements, workflow and schedule logs, and invalid submissions. These expose intermediate values, failed operations and validation messages.

Confirm the example defects in Table 6.2 against actual development records.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Stage 1 warning e-mail sent on every monitor run	Notification not gated by status	Send only when status changes Pending → At Risk; Escalation_Stage kept	Closed
BUG-02	Deadlines one time-zone offset wrong	Application time zone left at default	Set application time zone to Asia/Kolkata	Closed
BUG-03	Uptime credit created twice when monthly close re-run	No duplicate check	Check Service_Credit for same agreement, period, breach type before insert	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[Name of user 1]	Incident Manager	[Feedback received]	[Change made]
[Name of user 2]	Vendor Representative (portal)	[Feedback received]	[Change made]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The deployment sequence is Development → Stage → Production, with sample-data testing, role checks and release notes at each publish. Rollback restores a previous version; incident SLA snapshots preserve recorded deadlines. Production connections and schedules require separate authorisation and activation.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed in Development / Stage	Yes
2	Sample/test data removed or isolated	Yes
3	Roles, permissions and users configured in Production	Yes
4	Connections (cliq_sla_conn, uptimerobot_conn) authorised in Production	Yes
5	Schedules and workflows enabled; Scheduler_Control.Next_Run initialised	Yes
6	Application published and URL verified	Yes
7	Version / release notes recorded	Yes

7.2 Performance Optimisation

Table 7.2 summarises optimisations for the frequently executed monitor and reports. Query-count comparisons describe the code path rather than measured network latency.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Escalation policies loaded once into a Map instead of fetched per incident	DS-06	1 + n fetches per run	2 fetches per run	n fewer queries (n = open incidents)
Criteria fetch of open incidents only (not all incidents then filter)	DS-06	All incidents scanned	Open incidents only	Work proportional to open incidents
Notifications only on status change	DS-06, DS-05	Up to 4 e-mails/hour per incident	At most 1 per stage	No repeated mail; fewer sendmail calls
Default filter on Open Incidents report	Report	All incidents loaded	Only unresolved	Smaller result set

Optimisation Applied	Component	Before	After	Improvement
Report columns limited to 8; detail view for the rest	Reports	20+ columns	8 columns	Faster rendering, esp. mobile
Duplicate check before inserting credits/scorecards (upsert)	DS-08, DS-09	Re-run duplicated records	Idempotent	Safe re-execution

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role permission profiles	All forms, reports and pages	Unauthorised actions (e.g. Incident Manager editing contracts)
Record-level portal criteria	Incident, Agreement, Downtime, Credit, Scorecard	One vendor viewing another vendor's data
Read-only computed SLA fields (field rules)	Incident deadlines, applied targets, statuses	Manipulation of deadlines or SLA results
Blueprint-restricted transitions	Incident lifecycle	Skipping stages; back-dating resolution
Encrypted field	Monthly_Service_Fee, Credit_Amount	Exposure of commercial terms
OAuth / API-key connections (no keys in code)	Zoho Cliq, UptimeRobot	Credential leakage through scripts
Approval process for credits	Service_Credit	Unreviewed financial claims
Audit trail enabled	All forms	Untraceable changes; repudiation
Two-factor authentication for admin account	Zoho account	Account takeover

7.4 Governance and Administration

Governance combines platform audit history with escalation and credit records. Naming conventions use DS-nn script IDs, the sla function namespace and _conn connections. Backups comprise application-definition exports after releases and monthly data exports. Administrators maintain SLA parameters; developers maintain scripts and structure through the deployment environments.

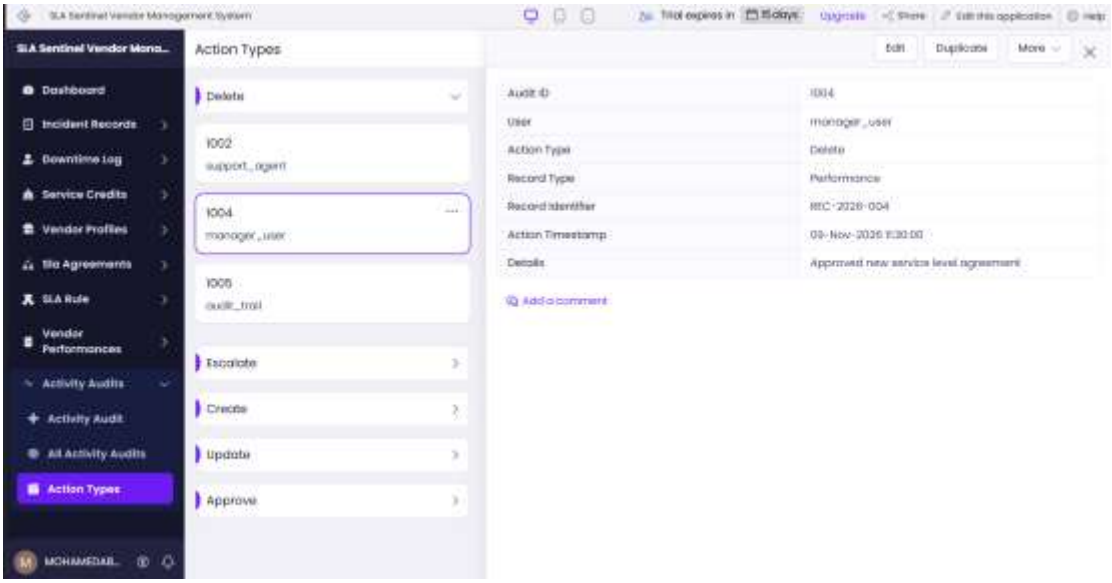


Figure 7.1: Activity Audit Record in the Action Types View

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The example scenario logs a Critical outage at 10:00 with response and resolution deadlines of 10:15 and 14:00. Acknowledgement at 10:10 meets response; restoration at 15:00 breaches resolution and produces 300 minutes of downtime. Scheduled warnings and escalation support follow-up. Monthly processing evaluates uptime, generates applicable credits for approval and updates the vendor scorecard.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Normalised data model and forms	Sections 4.2, 4.3, 5.2	Ten forms, two subforms, lookups; TC-01 to TC-03
2. Deadline calculation, independent breach detection, staged escalation (≤ 15 min)	Sections 5.5, 5.6 (DS-03 to DS-06, DS-10)	TC-07 to TC-11; Escalation Log
3. Monthly uptime and service credits with approval	Section 5.6 (DS-08, DS-11)	TC-13, TC-14
4. Role-based dashboards, reports and scorecards	Sections 5.3, 5.4, 5.9 (DS-09)	Thirteen reports, three pages; TC-15
5. Integration with Zoho Mail, Zoho Cliq and external API	Section 5.7	TC-11 (Cliq), DS-13 sync, Postman test
6. Secure production deployment with governance	Chapter 7, Section 5.8	TC-16, TC-17; deployment checklist

8.3 Comparison with the Existing System

Table 8.2 compares the application with estimated manual-process performance.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Finding the applicable SLA deadline	Look up PDF contract (~5 min per incident)	Computed on save (0 min)	Fully automated
Breach detection	At month-end review, if at all	Within 15 minutes	From weeks to minutes
Escalation	Ad hoc phone calls	Four staged, logged escalations	Consistent and auditable
Uptime calculation	Manual, approximate, maintenance not separated	Automatic monthly, planned maintenance excluded	Accurate and repeatable
Service-credit claims	Rarely calculated	Generated from rules and approved	Credits no longer missed
Vendor comparison	Not available	Monthly weighted scorecard and charts	Objective, historical
Data access	Shared spreadsheet, no access control	Role-based with vendor portal and audit trail	Secure and traceable

8.4 Limitations of the Low-Code Approach

The 15-minute polling interval can delay detection; SLA clocks do not pause for business hours or holidays. Deluge execution limits favour filtered queries and policy caching. Live countdowns require custom UI components. Data can be exported, but Creator-specific forms, Blueprints and Deluge logic would need rebuilding on another platform [5].

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code configuration covers forms, reports, permissions, Blueprints and approvals. Deluge supplies deadline, escalation, credit and scorecard logic. Pro-code becomes useful for tighter monitoring intervals, complex business calendars or richer interfaces [6].

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

SLA Sentinel centralises vendor contracts and incidents, snapshots applicable targets and evaluates response and resolution independently. Its monitoring, downtime and escalation records support contractual accountability.

Monthly credits and scorecards, dashboards, role-based access and integrations complete the design. The project demonstrates how visual configuration and Deluge can support SLA management; Chapter 6 identifies the test records requiring confirmation.

9.2 Future Enhancements

- **Business-hours SLA clocks** with working-hour calendars and public holidays per agreement.
- **Breach-risk prediction** using a Zia prediction model trained on historical incidents (severity, vendor, category, time of day).
- **Near-real-time monitoring** of Critical incidents through a Zoho Catalyst cron function running every minute.
- **Accounting integration** to apply approved service credits automatically as deductions in Zoho Books invoices.
- **Automatic incident creation** from monitoring alerts through a webhook, and two-way synchronisation with vendors' ticketing systems.
- **Live countdown widget** on the Incident Monitor page using a JavaScript widget.

9.3 Learning Outcomes and Career Relevance

The project develops skills in data modelling, forms and reports, Deluge validation and calculations, workflow integration, permissions and deployment. It also identifies limits in scheduling, execution capacity and UI customisation, clarifying when a low-code solution needs pro-code extensions.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	SLA Sentinel – 7 sections	5.1	Figure 5.1, Figure 4.5
II	Forms with ≥ 8 distinct field types	Y	Vendor, SLA_Agreement, Incident (15+ types)	5.2	Table 4.2–Table 4.6
II	Lookup field (relationship between forms)	Y	Incident → Vendor, SLA_Agreement	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	Y	SLA_Targets, Credit_Rules	4.2, 5.2	Table 4.3
II	Field validations / layout rules	Y	DS-02, DS-07; field rules on Incident	5.2	Appendix C
II	List report with filter, grouping, custom action	Y	Open Incidents	5.3	Figure 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Incident Board, Deadline Calendar, Incident Timeline	5.3	Table 5.1
II	Pivot table / chart report	Y	Incident Pivot; Breaches by Vendor; Uptime vs Target	5.3, 5.9	Table 5.1
II	Page / dashboard built with UI Builder	Y	SLA Command Centre, Incident Monitor, Vendor Performance Analytics	5.4	Figure 5.1
II	Deluge – form events (validate / success / field actions)	Y	DS-01, DS-02, DS-03, DS-07	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	Y	DS-04, DS-05, DS-09	5.5	Code 5.2
II	Email / in-app notifications	Y	DS-03, DS-05, DS-11, DS-12	5.5, 5.6	Appendix C
II	Form workflow	Y	5 form workflows	5.6	Table 5.3
II	Scheduled workflow	Y	SLA Monitor, Monthly Close, Housekeeping, Uptime Sync	5.6	Code 5.3
II	Approval process	Y	Service Credit Approval	5.6	DS-11
II	Blueprint	Y	Incident Lifecycle	4.4, 5.6	Figure 4.4, Figure 4.4 and Section 5.6
II	Batch / report workflow	Y	Recalculate SLA (multi-record custom action)	5.6	Table 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Testing & debugging evidence	Y	18 test cases, defect log	6	Table 6.1, Section 6.3
III	Integration with a Zoho service	Y	Zoho Mail, Zoho Cliq	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	Y	UptimeRobot getMonitors (DS-13)	5.7	Code 5.4
III	Webhook or Creator REST API	Y	Creator REST API v2 – Add Records (Incident)	5.7	Section 5.7 and Code 5.4
III	Users, roles and permission profiles	Y	3 roles / profiles	5.8	Section 5.8 and Table 4.7
III	Record-level access criteria / customer portal	Y	Vendor portal, own-vendor criteria	5.8	Table 4.7
III	Zia / AI feature	Y	Keyword extraction on Incident description	5.9	Figure 5.8
III	Analytics (charts / Zoho Analytics)	Y	Charts, pivot, scorecards	5.9	Figure 5.8
IV	Performance optimisation applied & measured	Y	DS-06 policy cache, report filters	7.2	Table 7.2
IV	Deployment across environments	Y	Development → Stage → Production	7.1	Section 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Encrypted fee fields, OAuth connections, profiles	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Audit trail, naming conventions, backups	7.4	Figure 7.1
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified; REST APIs used	5.10	—
V	Low-code limitations & spectrum analysis	Y	—	8.4, 8.5	—
—	Mobile app / portal access	Y	Creator mobile app, vendor portal	5.11	Section 5.11

Total features implemented: 31 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 DS-01 – Incident – Vendor field – On User Input

```
// Script ID : DS-01
// Location  : Form Incident -> Field Vendor -> On User Input
// Purpose   : Restrict the SLA Agreement lookup to the selected vendor's active
agreements
clear SLA_Agreement;
if(input.Vendor != null)
{
  agrIds = SLA_Agreement[Vendor == input.Vendor && SLA_Status ==
"Active"].ID.getAll();
  SLA_Agreement:ui.add(agrIds);
  if(agrIds.size() == 1)
  {
    input.SLA_Agreement = agrIds.get(0);
  }
}
```

C.2 DS-02 – Incident – On Add – On Validate

```
// Script ID : DS-02
// Location  : Form Incident -> On Add -> On Validate
// Purpose   : Block inconsistent incidents before they are saved
agr = SLA_Agreement[ID == input.SLA_Agreement];
if(agr.count() == 0)
{
  alert "Please select a valid SLA agreement.";
  cancel submit;
}
if(agr.Vendor != input.Vendor)
{
  alert "The selected agreement does not belong to the selected vendor.";
  cancel submit;
}
reportedOn = input.Reported_Date.toDate();
if(agr.SLA_Status != "Active" || reportedOn < agr.Start_Date || reportedOn >
agr.End_Date)
{
  alert "Agreement " + agr.SLA_ID + " is not active on the reported date.";
  cancel submit;
}
if(input.Detected_Date != null && input.Detected_Date > input.Reported_Date)
{
  alert "Detected time cannot be later than the reported time.";
  cancel submit;
}
if(input.Reported_Date > zoho.currenttime)
{
  alert "Reported time cannot be in the future.";
  cancel submit;
}
```

C.3 DS-03 – Incident – On Add – On Success

```
// Script ID : DS-03
// Location  : Form Incident -> On Add -> On Success
// Purpose   : Snapshot SLA targets (so later contract edits do not change
//             history), compute response/resolution deadlines, notify vendor
```

```
agr = SLA_Agreement[ID == input.SLA_Agreement];
ven = Vendor[ID == input.Vendor];
respTarget = 0;
resTarget = 0;
for each t in agr.SLA_Targets
{
  if(t.Severity == input.Severity)
  {
    respTarget = t.Response_Target_Min;
    resTarget = t.Resolution_Target_Min;
  }
}
input.Escalation_Stage = 0;
if(respTarget == 0 || resTarget == 0)
{
  // No target defined for this severity -> flag for administrator review
  input.Response_SLA_Status = "Not Defined";
  input.Resolution_SLA_Status = "Not Defined";
  sendmail
  [
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"SLA target missing for " + input.Severity + " on " + agr.SLA_ID
    message :"Incident " + input.Incident_ID + " could not be timed because the
agreement has no " + input.Severity + " target."
  ]
}
else
{
  input.Applied_Response_Target_Min = respTarget;
  input.Applied_Resolution_Target_Min = resTarget;
  input.Applied_Uptime_Target = agr.Guaranteed_Uptime;
  input.Response_Deadline = input.Reported_Date.addMinutes(respTarget);
  input.Resolution_Deadline = input.Reported_Date.addMinutes(resTarget);
  input.Response_SLA_Status = "Pending";
  input.Resolution_SLA_Status = "Pending";
  sendmail
  [
    from :zoho.adminuserid
    to :ven.Email
    subject :"[" + input.Severity + "] New incident " + input.Incident_ID + " - "
+ input.Incident_Title
    message :"A new incident has been logged against " + agr.Agreement_Name +
".<br>Acknowledge by: " + input.Response_Deadline + "<br>Resolve by: " +
input.Resolution_Deadline
  ]
}
```

C.4 DS-04 – Function sla.evaluateIncidentSLA

```
// Script ID : DS-04
// Location  : Function sla.evaluateIncidentSLA(int incId)
//             Called from Incident -> On Edit -> On Success, Blueprint
transitions
//             and the "Recalculate SLA" custom action
// Purpose   : Evaluate response and resolution SLAs independently
void sla.evaluateIncidentSLA(int incId)
{
  inc = Incident[ID == incId];
  if(inc.Response_SLA_Status == "Not Defined")
  {
    return;
  }
```

```
  openStates = {"Pending","At Risk"};
  // ---- Response SLA: the first acknowledgement counts as the response ----
  firstResponse = ifnull(inc.Acknowledged_At,inc.Responded_At);
  if(firstResponse != null && openStates.contains(inc.Response_SLA_Status))
  {
    if(firstResponse <= inc.Response_Deadline)
    {
      inc.Response_SLA_Status = "Met";
    }
    else
    {
      inc.Response_SLA_Status = "Breached";
    }
  }
  // ---- Resolution SLA ----
  if(inc.Resolved_At != null)
  {
    inc.Resolution_Duration_Min = ((inc.Resolved_At.toLong() -
inc.Reported_Date.toLong()) / 60000).round(0);
    if(openStates.contains(inc.Resolution_SLA_Status))
    {
      if(inc.Resolved_At <= inc.Resolution_Deadline)
      {
        inc.Resolution_SLA_Status = "Met";
      }
      else
      {
        inc.Resolution_SLA_Status = "Breached";
      }
    }
    // ---- Outage incidents produce exactly one downtime record ----
    if(inc.Service_Outage == true && Downtime_Log[Incident == incId].count() ==
0)
    {
      outStart = ifnull(inc.Detected_Date,inc.Reported_Date);
      mins = ((inc.Resolved_At.toLong() - outStart.toLong()) / 60000).round(0);
      insert into Downtime_Log
      [
        Added_User = zoho.loginuser
        Vendor = inc.Vendor
        SLA_Agreement = inc.SLA_Agreement
        Incident = incId
        Outage_Start = outStart
        Outage_End = inc.Resolved_At
        Duration_Min = mins
        Planned_Maintenance = false
      ];
      inc.Downtime_Min = mins;
    }
  }
}
```

C.5 DS-05 – Function sla.logEscalation

```
// Script ID : DS-05
// Location  : Function sla.logEscalation(int incId, int stage, string
breachType, string reason)
// Purpose   : Single place that records and sends every escalation
void sla.logEscalation(int incId, int stage, string breachType, string reason)
{
  inc = Incident[ID == incId];
  ven = Vendor[ID == inc.Vendor];
  pol = Escalation_Policy[Severity == inc.Severity];
```

```
  if(stage == 1)
  {
    recipient = ven.Email;
  }
  else if(stage == 2)
  {
    recipient = pol.Stage2_Recipient_Email;
  }
  else
  {
    recipient = pol.Stage3_Recipient_Email;
  }
  insert into Escalation_Log
  [
    Added_User = zoho.adminuser
    Incident = incId
    Stage = stage
    Breach_Type = breachType
    Recipient = recipient
    Reason = reason
    Escalated_On = zoho.currenttime
  ];
  sendmail
  [
    from :zoho.adminuserid
    to :recipient
    subject :"SLA " + breachType + " alert (Stage " + stage + ") - " +
inc.Incident_ID + " [" + inc.Severity + "]"
    message :reason + "<br><br>Vendor: " + ven.Vendor_Name + "<br>Response
deadline: " + inc.Response_Deadline + "<br>Resolution deadline: " +
inc.Resolution_Deadline
  ]
  if(inc.Severity == "Critical" && stage >= 2)
  {
    msg = Map();
    msg.put("text","CRITICAL SLA " + breachType + " breach: " + inc.Incident_ID +
" (" + ven.Vendor_Name + ") - " + reason);
    zoho.cliq.postToChannel("slaalerts",msg,"cliq_sla_conn");
  }
}
```

C.6 DS-06 – Schedule – SLA Monitor (every 15 minutes)

```
// Script ID : DS-06
// Location  : Schedule "SLA Monitor" -> runs on Scheduler_Control.Next_Run
//             (self-rescheduling every 15 minutes)
// Purpose   : Continuous deadline monitoring and staged escalation
now = zoho.currenttime;
// Load all escalation policies once (avoids a fetch inside the loop)
policyMap = Map();
for each p in Escalation_Policy[ID != 0]
{

policyMap.put(p.Severity,{"warn":p.Warning_Percent,"delay":p.Stage3_Delay_Min});
}
openList = Incident[Incident_Status != "Resolved" && Incident_Status != "Closed"
&& Response_SLA_Status != "Not Defined"];
for each inc in openList
{
  pol = policyMap.get(inc.Severity);
  warnPct = 75;
  stage3Delay = 60;
  if(pol != null)
```

```
  {
    warnPct = ifnull(pol.get("warn"),75);
    stage3Delay = ifnull(pol.get("delay"),60);
  }
  elapsedMin = (now.toLong() - inc.Reported_Date.toLong()) / 60000;
  // ---------- Response SLA ----------
  if(inc.Acknowledged_At == null)
  {
    if(now > inc.Response_Deadline && inc.Response_SLA_Status != "Breached")
    {
      inc.Response_SLA_Status = "Breached";
      thisapp.sla.logEscalation(inc.ID,2,"Response","Response deadline " +
inc.Response_Deadline + " passed without acknowledgement.");
      if(inc.Escalation_Stage < 2)
      {
        inc.Escalation_Stage = 2;
      }
    }
    else if(inc.Response_SLA_Status == "Pending" && elapsedMin >=
inc.Applied_Response_Target_Min * warnPct / 100)
    {
      inc.Response_SLA_Status = "At Risk";
      thisapp.sla.logEscalation(inc.ID,1,"Response",warnPct + "% of the response
time has been used. Please acknowledge the incident.");
      if(inc.Escalation_Stage < 1)
      {
        inc.Escalation_Stage = 1;
      }
    }
  }
  // ---------- Resolution SLA ----------
  if(now > inc.Resolution_Deadline)
  {
    if(inc.Resolution_SLA_Status != "Breached")
    {
      inc.Resolution_SLA_Status = "Breached";
      thisapp.sla.logEscalation(inc.ID,2,"Resolution","Resolution deadline " +
inc.Resolution_Deadline + " has been exceeded.");
      if(inc.Escalation_Stage < 2)
      {
        inc.Escalation_Stage = 2;
      }
    }
    else if(inc.Escalation_Stage < 3 && now.toLong() -
inc.Resolution_Deadline.toLong() >= stage3Delay * 60000)
    {
      thisapp.sla.logEscalation(inc.ID,3,"Resolution","Breach has continued for
more than " + stage3Delay + " minutes. Escalated to the SLA Administrator.");
      inc.Escalation_Stage = 3;
    }
  }
  else if(inc.Resolution_SLA_Status == "Pending" && elapsedMin >=
inc.Applied_Resolution_Target_Min * warnPct / 100)
  {
    inc.Resolution_SLA_Status = "At Risk";
    thisapp.sla.logEscalation(inc.ID,1,"Resolution",warnPct + "% of the
resolution time has been used.");
    if(inc.Escalation_Stage < 1)
    {
      inc.Escalation_Stage = 1;
    }
  }
  inc.Last_Checked = now;
}
// Re-arm the schedule for the next run
```

```
ctl = Scheduler_Control[Job_Name == "SLA_MONITOR"];
ctl.Last_Run = now;
ctl.Next_Run = now.addMinutes(15);
```

C.7 DS-07 – Downtime_Log – On Add / On Edit – On Validate

```
// Script ID : DS-07
// Location  : Form Downtime_Log -> On Add & On Edit -> On Validate
// Purpose   : Validate outage window and compute duration
if(input.Outage_End == null || input.Outage_End <= input.Outage_Start)
{
  alert "Outage end must be later than outage start.";
  cancel submit;
}
input.Duration_Min = ((input.Outage_End.toLong() - input.Outage_Start.toLong()) /
60000).round(0);
// Overlapping records for the same agreement would double-count downtime
overlap = Downtime_Log[SLA_Agreement == input.SLA_Agreement && ID != input.ID &&
Outage_Start < input.Outage_End && Outage_End > input.Outage_Start].count();
if(overlap > 0)
{
  alert "This outage overlaps an existing downtime record for the same
agreement.";
  cancel submit;
}
```

C.8 DS-08 – Schedule – Monthly SLA Close

```
// Script ID : DS-08
// Location  : Schedule "Monthly SLA Close" -> 1st of every month, 02:00
// Purpose   : Uptime %, service credits and vendor scorecards for last month
today = zoho.currentdate;
periodStart = today.subMonth(1).toStartOfMonth();
nextStart = today.toStartOfMonth();
startDT = (periodStart.toString("dd-MMM-yyyy") + " 00:00:00").toDateTime();
endDT = (nextStart.toString("dd-MMM-yyyy") + " 00:00:00").toDateTime();
totalMin = periodStart.daysBetween(nextStart) * 24 * 60;
periodLabel = periodStart.toString("MMM-yyyy");
for each agr in SLA_Agreement[SLA_Status == "Active"]
{
  // ---- 1. Achieved uptime (planned maintenance excluded) ----
  downMin = 0;
  for each d in Downtime_Log[SLA_Agreement == agr.ID && Planned_Maintenance ==
false && Outage_Start >= startDT && Outage_Start < endDT]
  {
    downMin = downMin + d.Duration_Min;
  }
  uptime = ((totalMin - downMin) * 100.0 / totalMin).round(3);
  agr.Last_Period = periodLabel;
  agr.Last_Period_Uptime = uptime;
  // ---- 2. Uptime credit: highest matching tier ----
  upPct = 0;
  resBreaches = Incident[SLA_Agreement == agr.ID && Resolution_SLA_Status ==
"Breached" && Reported_Date >= startDT && Reported_Date < endDT].count();
  resPct = 0;
  for each rule in agr.Credit_Rules
  {
    if(rule.Breach_Type == "Uptime" && uptime < rule.Threshold &&
rule.Credit_Percent > upPct)
    {
      upPct = rule.Credit_Percent;
    }
```

```
    if(rule.Breach_Type == "Resolution" && resBreaches > 0)
    {
      // Credit_Percent per breach, Threshold = cap (%)
      resPct = resBreaches * rule.Credit_Percent;
      if(resPct > rule.Threshold)
      {
        resPct = rule.Threshold;
      }
    }
  }
  creditInputs = {"Uptime":upPct,"Resolution":resPct};
  measured = {"Uptime":uptime,"Resolution":resBreaches};
  for each bType in creditInputs.keys()
  {
    pct = creditInputs.get(bType);
    exists = Service_Credit[SLA_Agreement == agr.ID && Period == periodLabel &&
Breach_Type == bType].count();
    if(pct > 0 && exists == 0)
    {
      insert into Service_Credit
      [
        Added_User = zoho.adminuser
        Vendor = agr.Vendor
        SLA_Agreement = agr.ID
        Period = periodLabel
        Breach_Type = bType
        Measured_Value = measured.get(bType)
        Credit_Percent = pct
        Credit_Amount = (agr.Monthly_Service_Fee * pct / 100).round(2)
        Approval_Status = "Pending Approval"
      ];
    }
  }
}
// ---- 3. Vendor scorecards ----
for each v in Vendor[Vendor_Status == "Active"]
{
  thisapp.sla.generateScorecard(v.ID,periodLabel,startDT,endDT);
}
```

C.9 DS-09 – Function sla.generateScorecard

```
// Script ID : DS-09
// Location  : Function sla.generateScorecard(int vendorId, string periodLabel,
//                                            datetime startDT, datetime endDT)
// Purpose   : Weighted vendor performance score for one period
void sla.generateScorecard(int vendorId, string periodLabel, datetime startDT,
datetime endDT)
{
  total = 0;
  respMet = 0;
  respEval = 0;
  resMet = 0;
  resEval = 0;
  breaches = 0;
  for each i in Incident[Vendor == vendorId && Reported_Date >= startDT &&
Reported_Date < endDT]
  {
    total = total + 1;
    if(i.Response_SLA_Status == "Met")
    {
      respMet = respMet + 1;
      respEval = respEval + 1;
```

```
    }
    else if(i.Response_SLA_Status == "Breached")
    {
      respEval = respEval + 1;
      breaches = breaches + 1;
    }
    if(i.Resolution_SLA_Status == "Met")
    {
      resMet = resMet + 1;
      resEval = resEval + 1;
    }
    else if(i.Resolution_SLA_Status == "Breached")
    {
      resEval = resEval + 1;
      breaches = breaches + 1;
    }
  }
  respPct = 100.0;
  if(respEval > 0)
  {
    respPct = (respMet * 100.0 / respEval).round(1);
  }
  resPct = 100.0;
  if(resEval > 0)
  {
    resPct = (resMet * 100.0 / resEval).round(1);
  }
  // Uptime score: 100 when target met, minus 1 point per 0.01 % shortfall
  upSum = 0.0;
  upCount = 0;
  upScoreSum = 0.0;
  for each a in SLA_Agreement[Vendor == vendorId && Last_Period == periodLabel]
  {
    upSum = upSum + a.Last_Period_Uptime;
    upCount = upCount + 1;
    gap = (a.Guaranteed_Uptime - a.Last_Period_Uptime) * 100;
    s = 100.0;
    if(gap > 0)
    {
      s = 100 - gap;
    }
    if(s < 0)
    {
      s = 0;
    }
    upScoreSum = upScoreSum + s;
  }
  avgUptime = 100.0;
  upScore = 100.0;
  if(upCount > 0)
  {
    avgUptime = (upSum / upCount).round(3);
    upScore = upScoreSum / upCount;
  }
  credits = 0.0;
  for each c in Service_Credit[Vendor == vendorId && Period == periodLabel]
  {
    credits = credits + c.Credit_Amount;
  }
  score = (0.3 * respPct + 0.4 * resPct + 0.3 * upScore).round(1);
  if(score >= 90)
  {
    rating = "Excellent";
  }
  else if(score >= 75)
```

```
  {
    rating = "Good";
  }
  else if(score >= 60)
  {
    rating = "Average";
  }
  else
  {
    rating = "Poor";
  }
  existing = Vendor_Scorecard[Vendor == vendorId && Period == periodLabel];
  if(existing.count() == 0)
  {
    insert into Vendor_Scorecard
    [
      Added_User = zoho.adminuser
      Vendor = vendorId
      Period = periodLabel
      Total_Incidents = total
      Response_Compliance = respPct
      Resolution_Compliance = resPct
      Achieved_Uptime = avgUptime
      Breach_Count = breaches
      Total_Credits = credits
      Performance_Score = score
      Rating = rating
    ];
  }
  else
  {
    existing.Total_Incidents = total;
    existing.Response_Compliance = respPct;
    existing.Resolution_Compliance = resPct;
    existing.Achieved_Uptime = avgUptime;
    existing.Breach_Count = breaches;
    existing.Total_Credits = credits;
    existing.Performance_Score = score;
    existing.Rating = rating;
  }
}
```

C.10 DS-10 – Blueprint – Incident Lifecycle transitions

```
// Script ID : DS-10
// Location  : Blueprint "Incident Lifecycle" -> After-transition scripts
// Purpose   : Stamp timestamps and re-evaluate the SLA on each transition

// --- Transition: Acknowledge (Reported -> Acknowledged) ---
if(input.Acknowledged_At == null)
{
  input.Acknowledged_At = zoho.currenttime;
}
input.Incident_Status = "Acknowledged";
thisapp.sla.evaluateIncidentSLA(input.ID);

// --- Transition: Start Work (Acknowledged -> In Progress) ---
if(input.Responded_At == null)
{
  input.Responded_At = zoho.currenttime;
}
input.Incident_Status = "In Progress";
```

```
// --- Transition: Resolve (In Progress -> Resolved) ---
// Resolution_Notes is a mandatory field in this transition's "During" form
if(input.Resolved_At == null)
{
  input.Resolved_At = zoho.currenttime;
}
input.Incident_Status = "Resolved";
thisapp.sla.evaluateIncidentSLA(input.ID);

// --- Transition: Reopen (Resolved -> Reopened), allowed within 72 hours ---
input.Incident_Status = "Reopened";
input.Reopen_Count = ifnull(input.Reopen_Count,0) + 1;
```

C.11 DS-11 – Service_Credit – Approval process

```
// Script ID : DS-11
// Location  : Approval process "Service Credit Approval" (form Service_Credit)
// Purpose   : Record the decision and inform the vendor

// --- On Approve ---
input.Approval_Status = "Approved";
input.Approved_By = zoho.loginuser;
input.Decision_On = zoho.currenttime;
ven = Vendor[ID == input.Vendor];
sendmail
[
  from :zoho.adminuserid
  to :ven.Email
  subject :"Service credit " + input.Credit_ID + " approved for " + input.Period
  message :"A service credit of INR " + input.Credit_Amount + " (" +
input.Credit_Percent + "% of the monthly fee) has been approved for the " +
input.Breach_Type + " SLA breach in " + input.Period + "."
]

// --- On Reject ---
input.Approval_Status = "Rejected";
input.Approved_By = zoho.loginuser;
input.Decision_On = zoho.currenttime;
```

C.12 DS-12 – Schedule – Daily Agreement Housekeeping

```
// Script ID : DS-12
// Location  : Schedule "Agreement Housekeeping" -> daily, 08:00
// Purpose   : Expire lapsed agreements, warn about upcoming expiries
today = zoho.currentdate;
for each a in SLA_Agreement[SLA_Status == "Active" && End_Date < today]
{
  a.SLA_Status = "Expired";
}
expiring = SLA_Agreement[SLA_Status == "Active" && End_Date >= today && End_Date
<= today.addDay(30)];
if(expiring.count() > 0)
{
  body = "The following SLA agreements expire within 30 days:<br>";
  for each a in expiring
  {
    body = body + a.SLA_ID + " - " + a.Agreement_Name + " (ends " + a.End_Date +
")<br>";
  }
  sendmail
```

```
  [
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"SLA agreements nearing expiry"
    message :body
  ]
}
```

C.13 DS-13 – Function sla.syncUptimeRobot

```
// Script ID : DS-13
// Location  : Function sla.syncUptimeRobot() -> called by daily schedule, 06:00
// Purpose   : Pull the independent 30-day uptime ratio of each monitored
//             service so it can be compared with downtime recorded in the app
void sla.syncUptimeRobot()
{
  params = Map();
  params.put("format","json");
  params.put("custom_uptime_ratios","30");
  response = invokeurl
  [
    url :"https://api.uptimerobot.com/v2/getMonitors"
    type :POST
    parameters:params
    connection:"uptimerobot_conn"
  ];
  info response;
  if(response.get("stat") != "ok")
  {
    return;
  }
  for each m in response.get("monitors")
  {
    agrList = SLA_Agreement[Monitor_ID == m.get("id").toString()];
    for each a in agrList
    {
      a.External_Uptime_30d = m.get("custom_uptime_ratio").toDecimal();
      a.External_Sync_On = zoho.currenttime;
    }
  }
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the SLA Sentinel published dashboard in Chrome, Edge or Firefox. On a phone, install the Zoho Creator app, sign in with the same account and open SLA Sentinel. Vendor representatives open the vendor portal link sent to them by e-mail and sign in with their portal credentials.

D.2 Instructions for the Administrator / SLA Manager

- Open **Vendors → Add Vendor**, enter the vendor's details and portal user e-mail, and submit.
- Open **Agreements → New SLA Agreement**, select the vendor, enter uptime, maintenance window and monthly fee, add one **SLA Targets** row per severity and the **Credit Rules** tiers, then submit.
- Open **Settings → Escalation Policy** once to set the warning percentage, Stage 3 delay and recipients for each severity.
- Review the **SLA Command Centre** daily; open **Monitoring → Breach Register** for details of any breach.
- At the start of each month open **Credits & Performance → Service Credits** and approve or reject each pending credit.
- Use **Vendor Scorecards** and **Vendor Performance Analytics** for monthly reviews; export reports as PDF when needed.

D.3 Instructions for the Incident Manager

- Open **Incidents → Report Incident**, select the vendor (the agreement list fills automatically), choose category and severity, tick *Service Outage* if the service was down, and submit. Note the deadlines shown in the success message.
- Track incidents on **Incidents → Incident Board**; use the Blueprint buttons (Start Work, Hold, Resume, Escalate) as the incident progresses.
- Act on Stage 2 e-mails promptly — they mean a deadline has been missed.
- Record planned maintenance or outages not linked to an incident through **Monitoring → Downtime Log**, ticking *Planned Maintenance* where it applies.

D.4 Instructions for the Vendor Representative

- Sign in to the vendor portal and open **My Vendor Incidents**; incidents nearest their deadline appear first.
- Open an incident and click **Acknowledge** as soon as you receive it, then **Start Work**.
- Post progress notes with **Add Update**; when service is restored click **Resolve** and enter the resolution notes (mandatory).
- View your monthly compliance and score under **My SLA Performance** and approved credits under **Approved Credits**.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Mohamed Abshar
Register Number	240701317
Project Title	SLA Sentinel – Intelligent Service Level Agreement Monitoring and Vendor Performance Management System
Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Mohamed Abshar
Register No.	240701317
Department / Section	Computer Science and Engineering
Date of Viva	
Application URL Verified	Yes / No
Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

INTERNAL EXAMINER	EXTERNAL EXAMINER
Name: _____	Name: _____